# LegatorOCR – User guide

Edition 0.2.0-pre.32 (trial version). Work in progress; this English edition is translated from the Romanian original. Where a caption says the interface is in Romanian, the screenshot shows the Romanian interface.

## 1. What LegatorOCR is

LegatorOCR turns scans or photos of pages into a clean PDF with searchable text. It solves the same problem as ScanTailor (splitting pages, deskewing, cropping, cleaning), but adds PDF export and OCR. It is written in Rust, with its own code, each stage being a textbook technique. It runs locally, with no account and without sending the pages over the internet. Licence: MIT.

There are two ways of working, with the same engines underneath:

- **The graphical interface** (`legatorocr-gui`): you add pages, see the original and the result side by side, correct page by page and export.
- **The command line** (`legatorocr`): for large batches, for automation and for options the interface does not show.

### What it does

- splits double-page sheets into two pages, straightens skewed pages and crops the margins;
- cleans the background, stains, punch holes, fingers on the edge and form lines;
- writes PDF (including PDF/A for archiving) or images, with OCR in Romanian and 7 other languages;
- also produces plain text (`.txt`) and hOCR, and can sign or date-stamp the PDF.

### What it does not do (or has not been proven to do)

- It is a trial version (0.2.0-pre.32). It has been run by the author and his assistant only on Linux; **Windows, macOS and arm64 processors have not been tried by people**.
- OCR accuracy was measured on 8 good pages from a single Romanian magazine, against a reference text produced by another program, not corrected by a person. Do not take it as a guarantee for other kinds of print.
- OCR can make mistakes: spot-check the important pages.

## 2. Installation and first run

Download the archive for your system from the releases page (Releases) of the `chiuta/LegatorOCR` repository. Each archive has a `.sha256` file for verification.

| System | File | Notes |
|---|---|---|
| Linux x86-64 or ARM64 | `.tar.gz` | needs OpenGL 2+ and X11 or Wayland |

| System | File | Notes |
|---|---|---|
| Windows x86-64 | `.zip` | needs OpenGL 2+; it may be missing in plain RDP sessions |
| macOS Apple-silicon or Intel | `.zip` with `LegatorOCR.app` | **unsigned**: at first launch right-click → Open, or `xattr -dr com.apple.quarantine LegatorOCR.app` |

You can also build from source: `cargo build --release` (requires Rust 1.95 or newer).

## OCR

The program has its own OCR engine, which works without installation. Optionally you can install Tesseract, a separate program, which LegatorOCR uses automatically when it is present and has the requested language:

- Linux (Debian/Ubuntu): `sudo apt install tesseract-ocr tesseract-ocr-ron`
- macOS: `brew install tesseract tesseract-lang`
- Windows: `winget install UB-Mannheim.TesseractOCR` and adding it to `PATH`

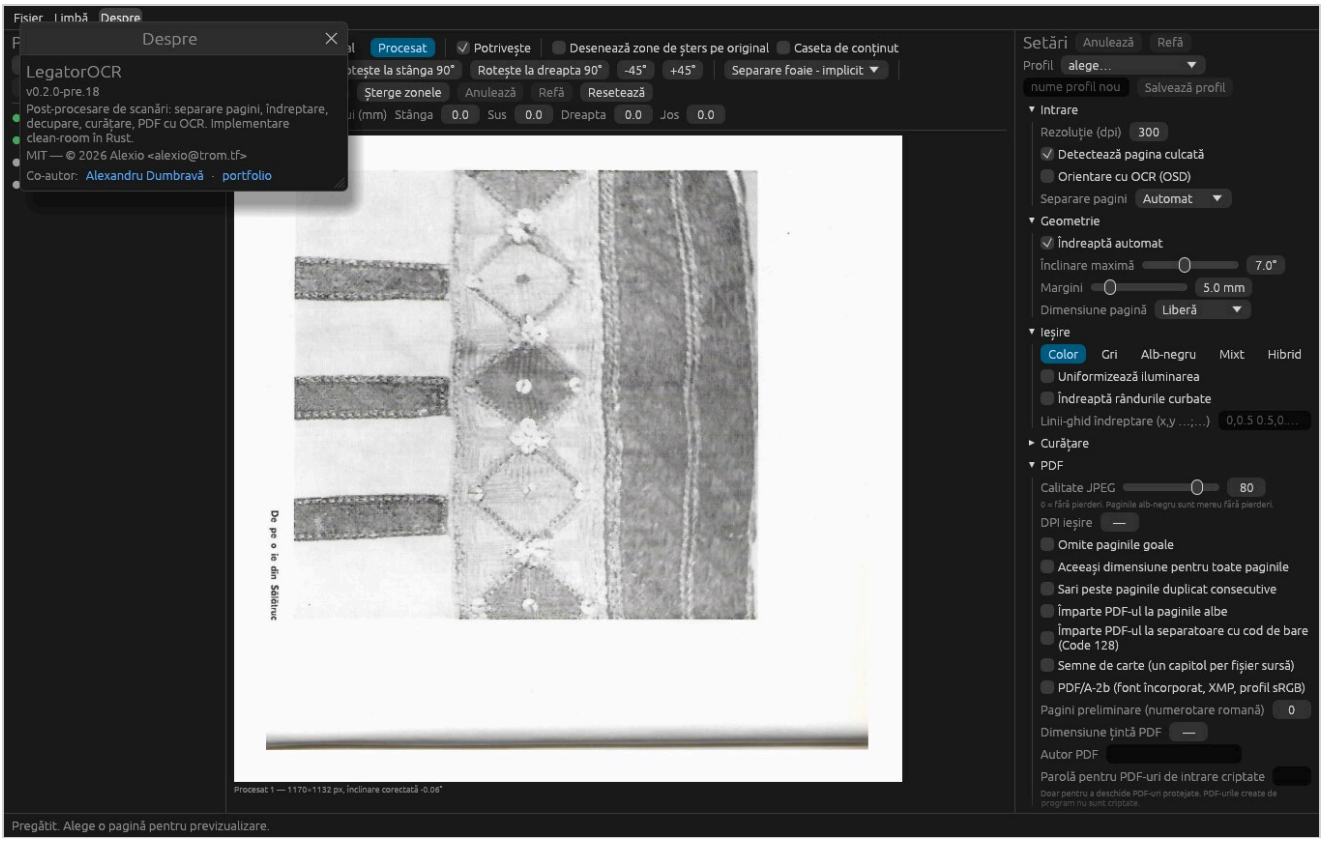
The Romanian model in the `apt` and `brew` packages is the small ("fast") variant. The "best" variant is downloaded manually and pointed to with `--tessdata DIR`. Which of them reads your pages better has not been measured.

## Check the installation

Run `legatorocr --doctor`. It prints the version, the native OCR engine, Tesseract and the language models, then auxiliary tools (`pdftoppm`, `gs`, `qpdf`, `mutool`). A "missing" result for an optional tool is not an error; the message below it says whether it is needed. If something does not work, copy the whole output of the command into a report (see section 10).

```
$ legatorocr --doctor
[ok     ] legatorocr                        0.2.0-pre.19
[ok     ] native OCR engine                 built in
[ok     ] tesseract                         tesseract 5.3.4
[ok     ] tesseract model ro (Romanian)     installed
[ok     ] tesseract model en (English)      installed
[ok     ] tesseract model fr (French)       installed
[ok     ] tesseract model it (Italian)      installed
[ok     ] tesseract model es (Spanish)      installed
[ok     ] tesseract model pt (Portuguese)   installed
[ok     ] tesseract model de (German)       installed
[ok     ] tesseract model hu (Hungarian)    installed
[ok     ] pdftoppm                          pdftoppm
[missing] mutool                            not found
          -> optional: alternative renderer for those cases
[ok     ] gs                                gs
[ok     ] qpdf                              qpdf
14 check(s) ok, 1 missing. Missing helpers are optional unless a hint says otherwise.
```

The output of the legatorocr --doctor command (pre.20, Linux, in a real terminal; the same for all terminal screenshots in the guide, made with xterm): "missing" for mutool is an optional tool, not an error.



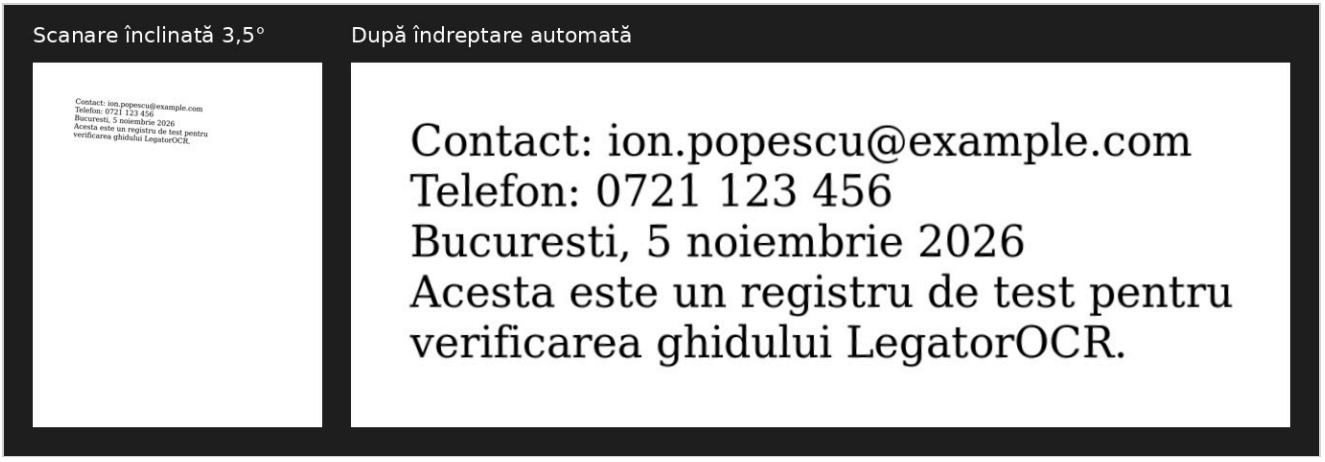

The About menu shows the installed version (here 0.2.0-pre.18). Screenshot from the pre.18 build on Linux, interface in Romanian.

## 3. The workflow, step by step

For each page, the program goes through the stages below in this order. The analysis is done on a copy of about 150 dpi, and the result is then applied to the scan at its full resolution.

1. **Orientation.** Pages lying on their side (a quarter turn) are detected from the letters of the page: a page with pictures or with few words is not rotated by this test. For the exact direction, Tesseract (OSD) or manual rotation can be used.
2. **Sheet splitting.** A landscape sheet with two pages is cut in the middle, at the empty band in the centre. It can be automatic, at the middle, or off.
3. **Deskewing.** The skew angle is estimated from the text lines and the page is rotated back. The angle can be fixed manually.
4. **Content box.** The area with text and images is found; scanner shadows and dust are ignored. Everything outside it is repainted in the paper colour.
5. **Margins and size.** Margins in millimetres, optionally a fixed page size or the same size for the whole book.
6. **Straightening curved lines** (optional). For dense book pages with lines curved near the spine.
7. **Lighting** (optional). Evens out an uneven background.

8. **Cleaning and binarization.** Removes stains, punch holes, fingers, lines; then black and white, grey, colour, mixed or hybrid.
9. **PDF and OCR.** The PDF (or the images) is written and, if ticked, the text layer.



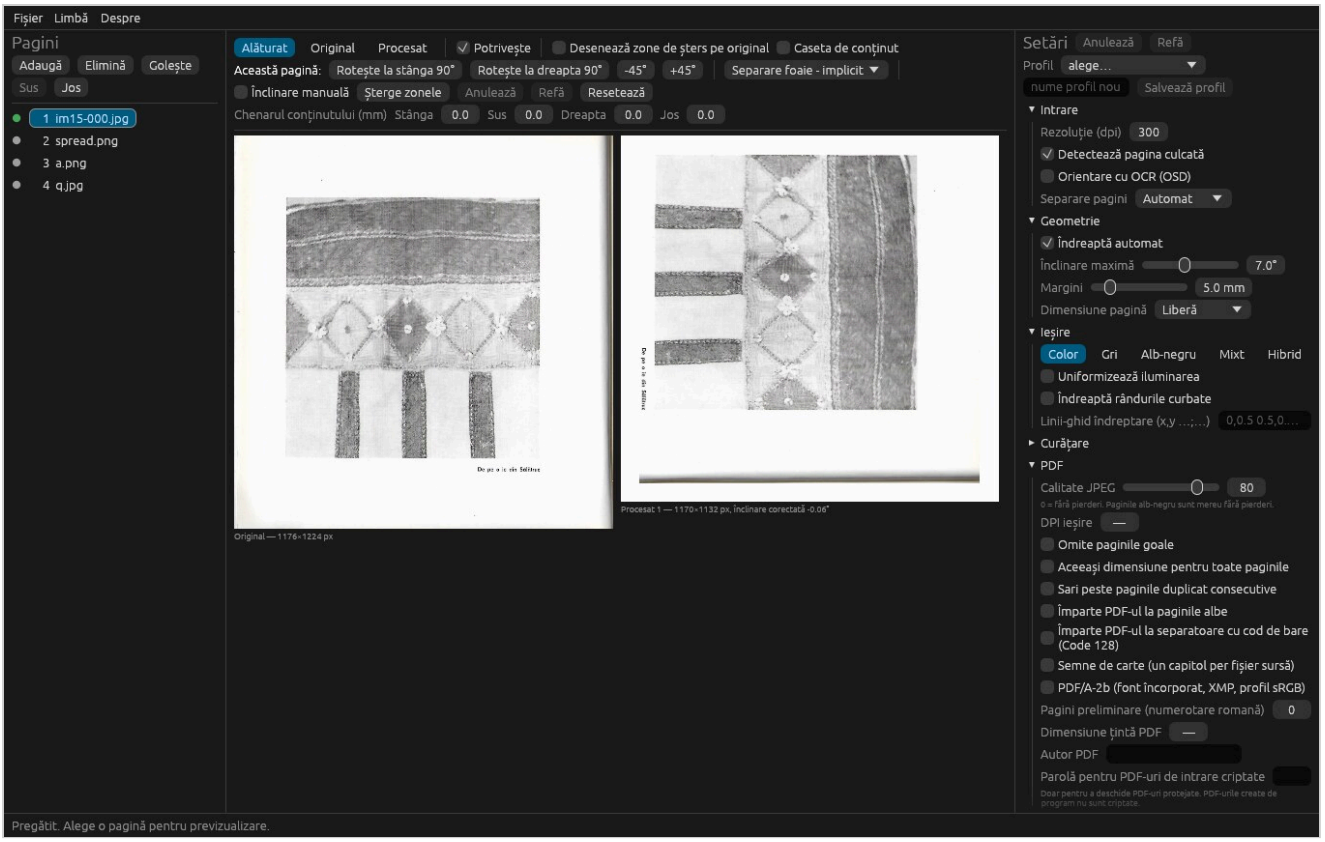

Automatic deskewing: a scan skewed by 3.5° (a test sheet generated by me) and the result; the program measured -3.50°.

### Working rules that save time

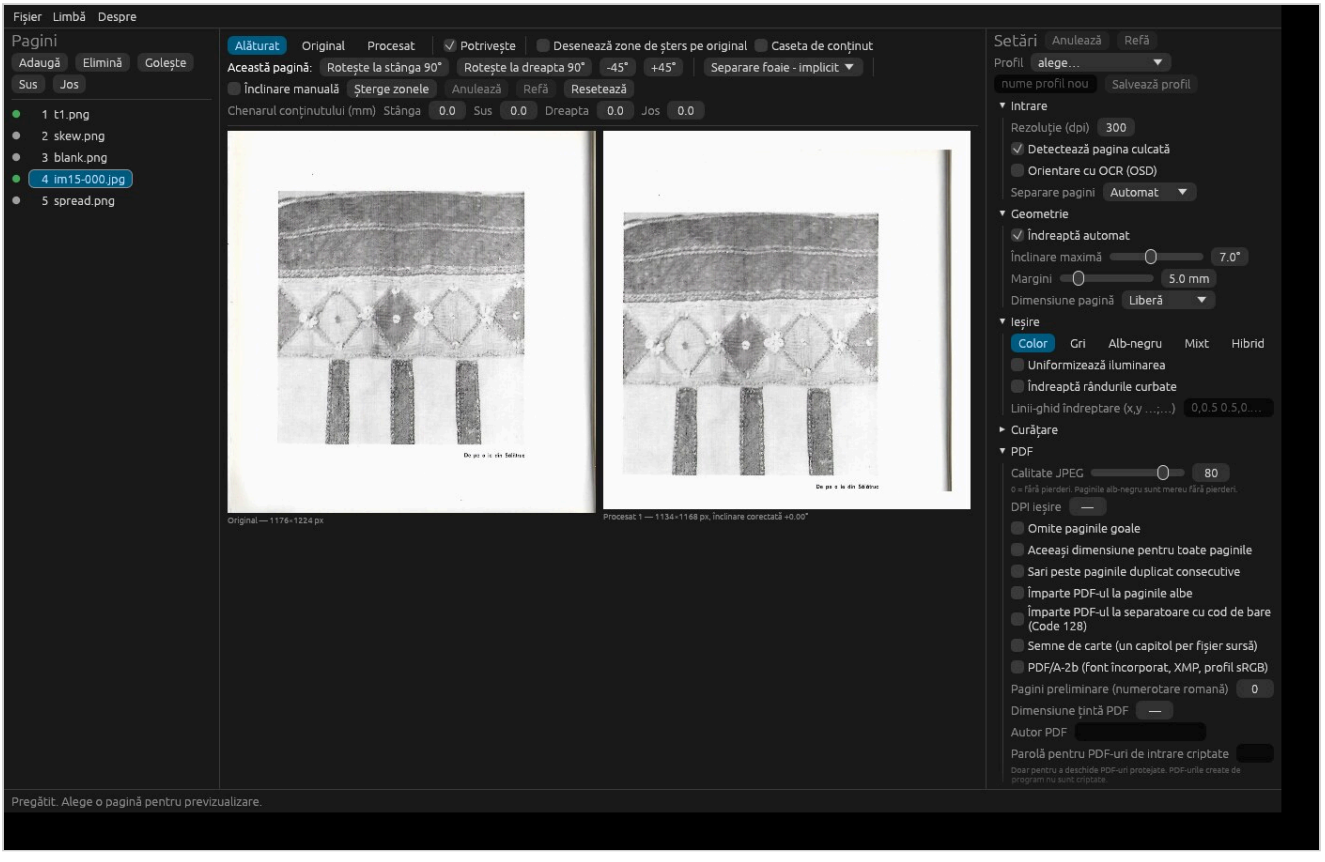
- First set the correct **resolution** (usually 300 dpi for text). The program reads it from the image if it is between 150 and 1200; otherwise it assumes 300.
- Choose a **profile** suited to the material (section 6) before touching individual settings.
- Process 10–20 pages as a trial, check the result, then start the large batch.
- The interface marks pages "to check" (blank page, borderline skew, nothing found, content stuck to the edge). Go through them with the "Next" button.

## 4. The graphical interface

The window has three areas: the page list on the left, the preview in the centre and the settings on the right. The **File** menu contains "Add images", "Add a folder", "Open project", "Save project", "Export PDF" and "Export images". The interface language (8 languages) is changed from the **Language** menu.



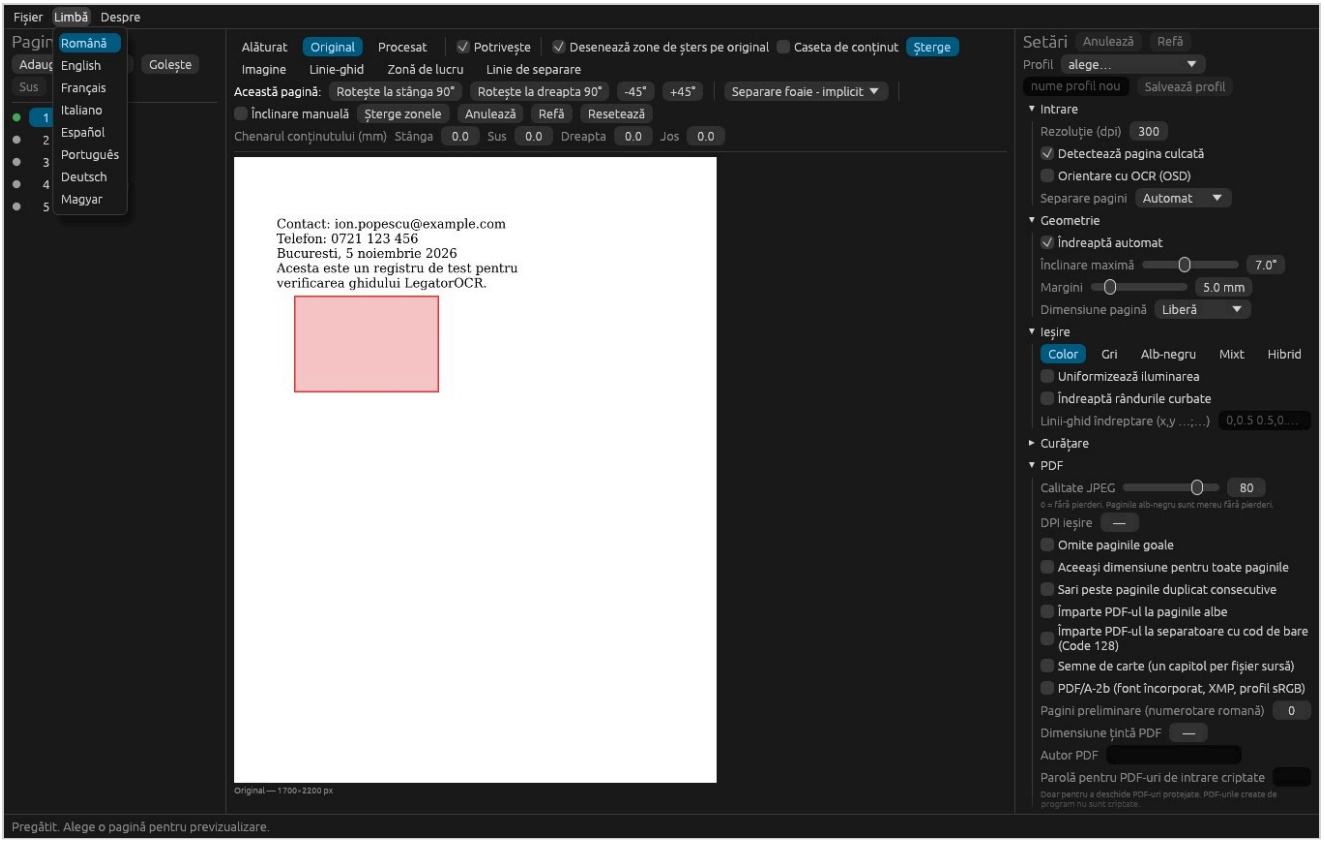

The main window: the page list on the left, the original and the processed page in the middle, the settings on the right. The example page is a photo without text, so the detection of a page lying on its side rotated it wrongly in pre.18 (fixed in pre.19, see below). Screenshot pre.18 on Linux.




The same page in pre.19, with "Detect page lying on its side" ticked: it is no longer rotated (processed 1134×1168 px, skew +0.00°). Screenshot pre.19 on Linux.

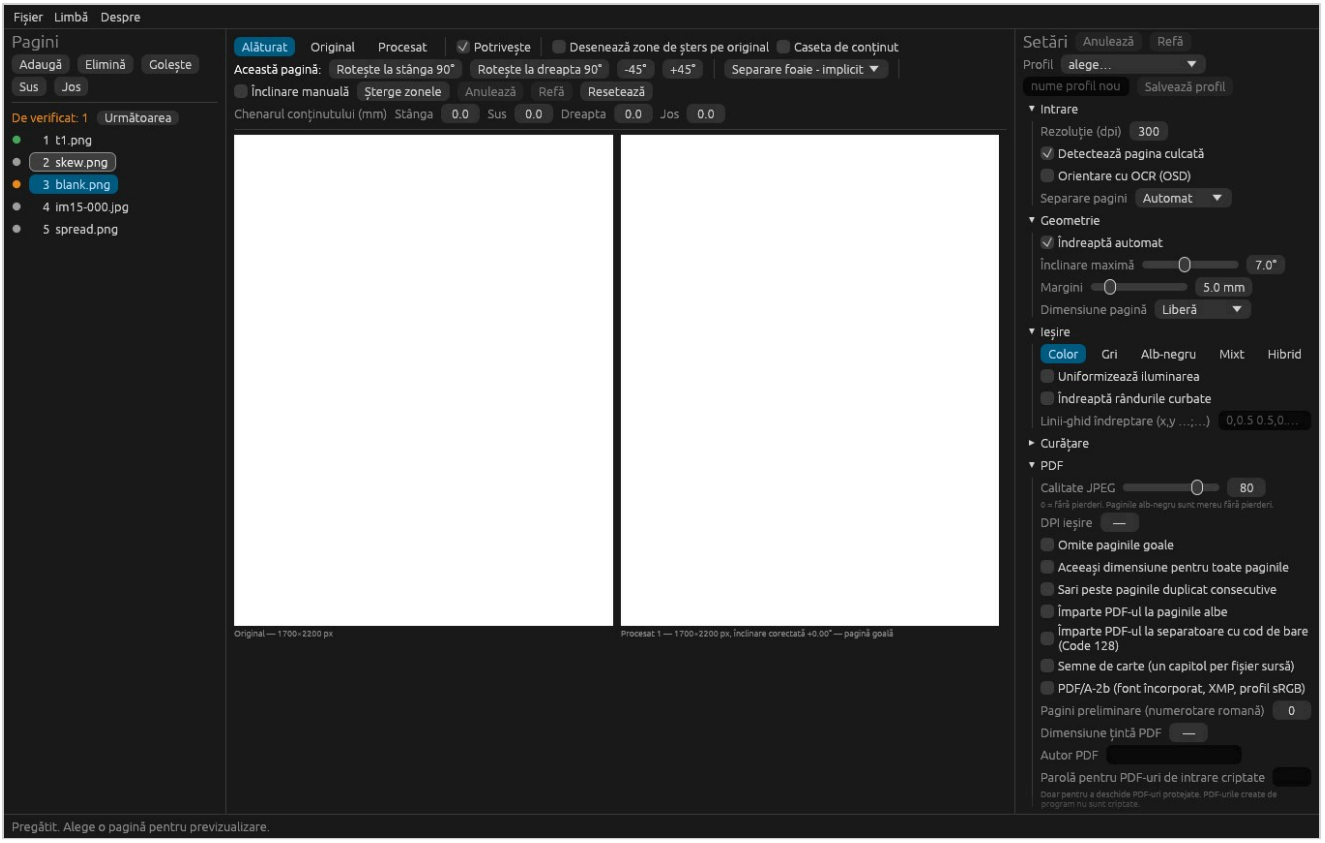



The File menu: adds images or a folder, opens and saves the project, exports PDF or images.




The Language menu: the eight interface languages (Romanian, English, French, Italian, Spanish, Portuguese, German, Hungarian).

## The page list

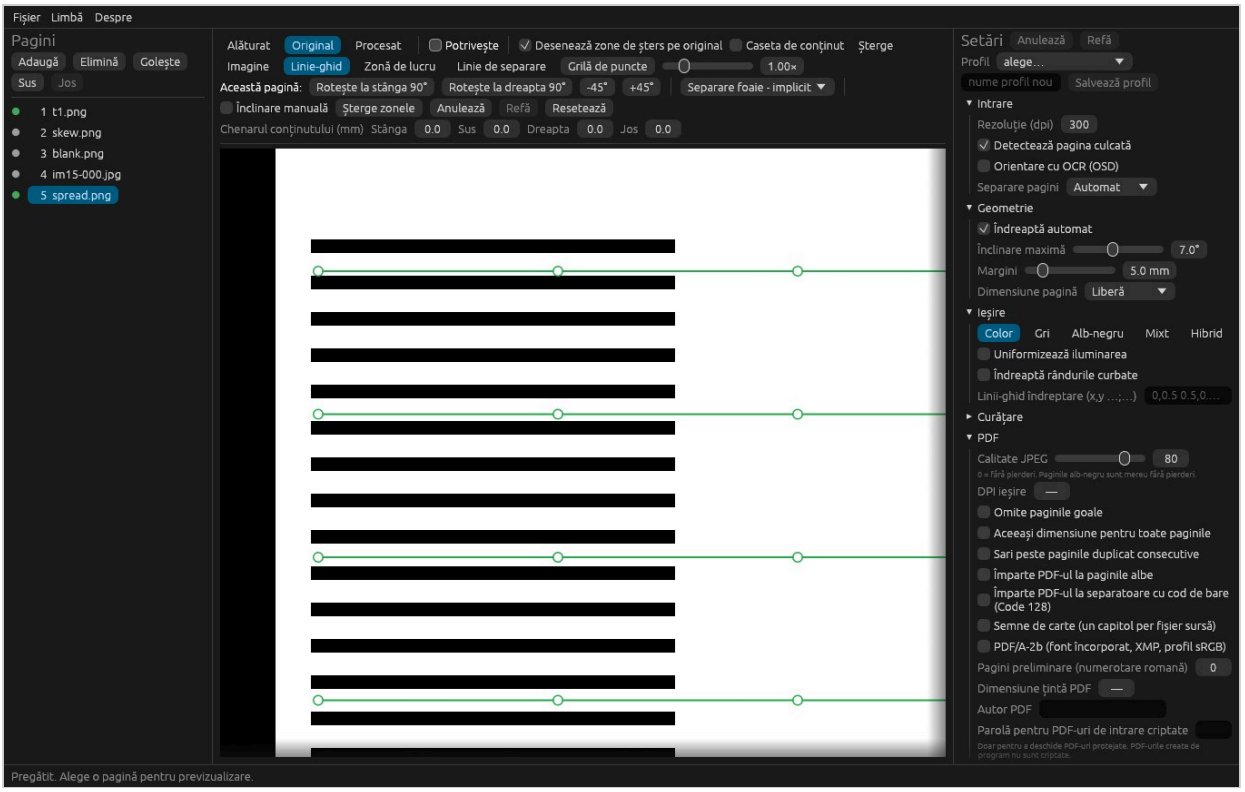



A completely white page is marked automatically: an orange dot in the list and the "To check: 1" panel with the "Next" button.

Drag images, PDFs, archives or a folder into it, or press "Add". The Up/Down buttons change the order, "Remove" takes out the chosen page, "Clear" removes everything. The order in the list is the order in the PDF.

## The preview

- **Side by side / Original / Processed** chooses what you see. Zones are drawn on *Original*; the content box is corrected on *Processed*.
- **Fit** enlarges or shrinks the image so that it fits. The zoom slider (0.2×–4×) appears only when *Fit* is unticked.

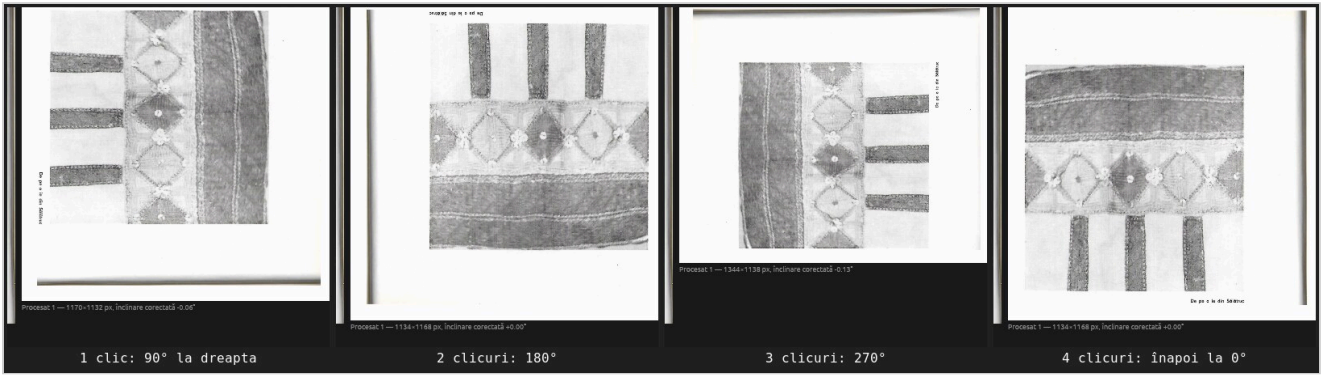



With "Fit" unticked the zoom slider appears (here 1.00×).

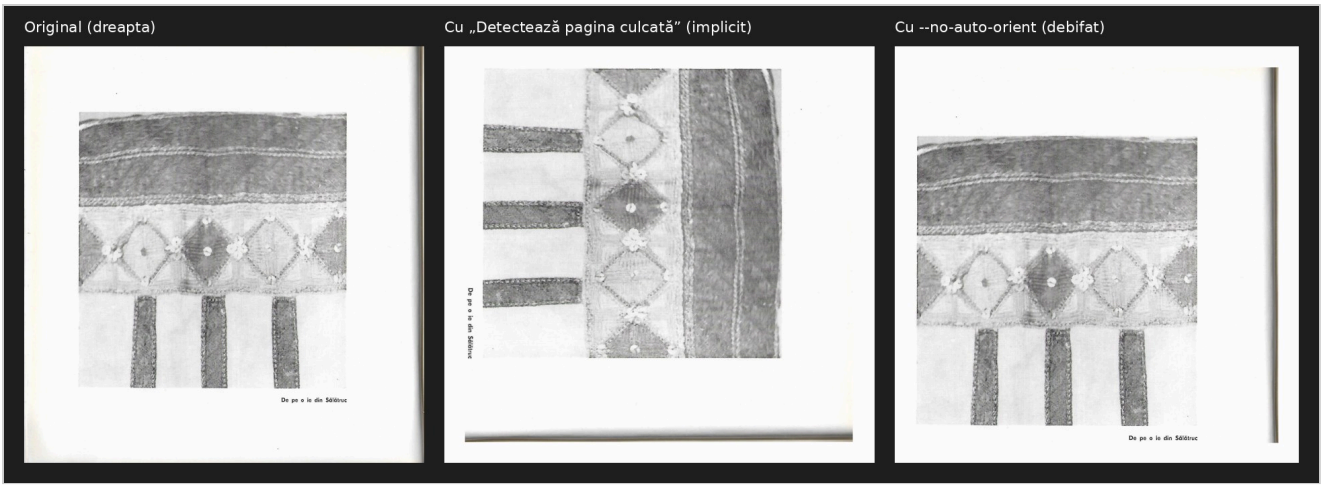
## The "This page" bar

Here you correct only the chosen page: rotation left or right by 90°, ±45° steps, manual skew, the sheet-splitting mode (default, automatic, middle, off), the content frame in millimetres (left, top, right, bottom; positive values enlarge it) and "Reset". **Undo** and **Redo** (Ctrl+Z, Ctrl+Y) work for all these commands, not only for zones.

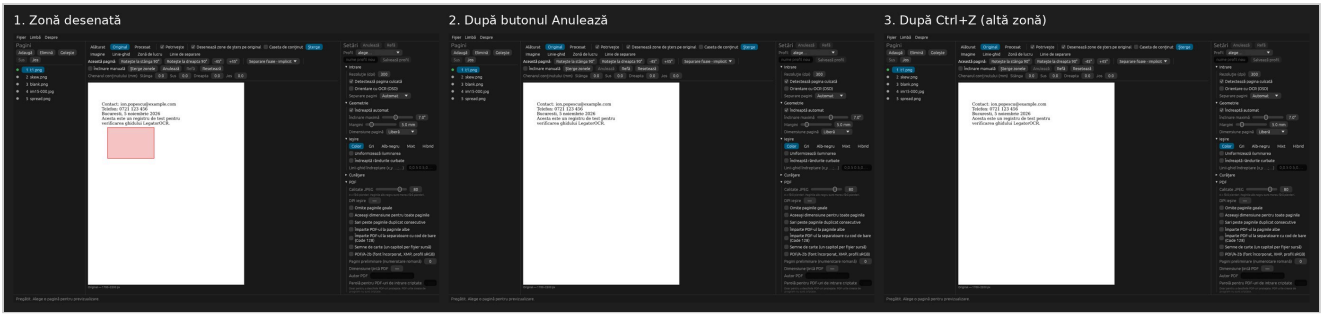
**Manual rotation locks the automatic orientation** (since pre.18): after you press a rotation button or ±45°, the page is no longer rotated by the detection of a page lying on its side or by OSD, not even at 0°. "Reset" gives the automatic orientation back. If a page with pictures is rotated wrongly from the start, press rotate four times (or untick "Orientation with OCR (OSD)" and "Detect page lying on its side" in the settings, under Input).




Four clicks in a row on "Rotate right 90°": 90°, 180°, 270°, then back to 0°, with no jumps. Screenshot pre.18 on Linux. I could not check on Windows 11.



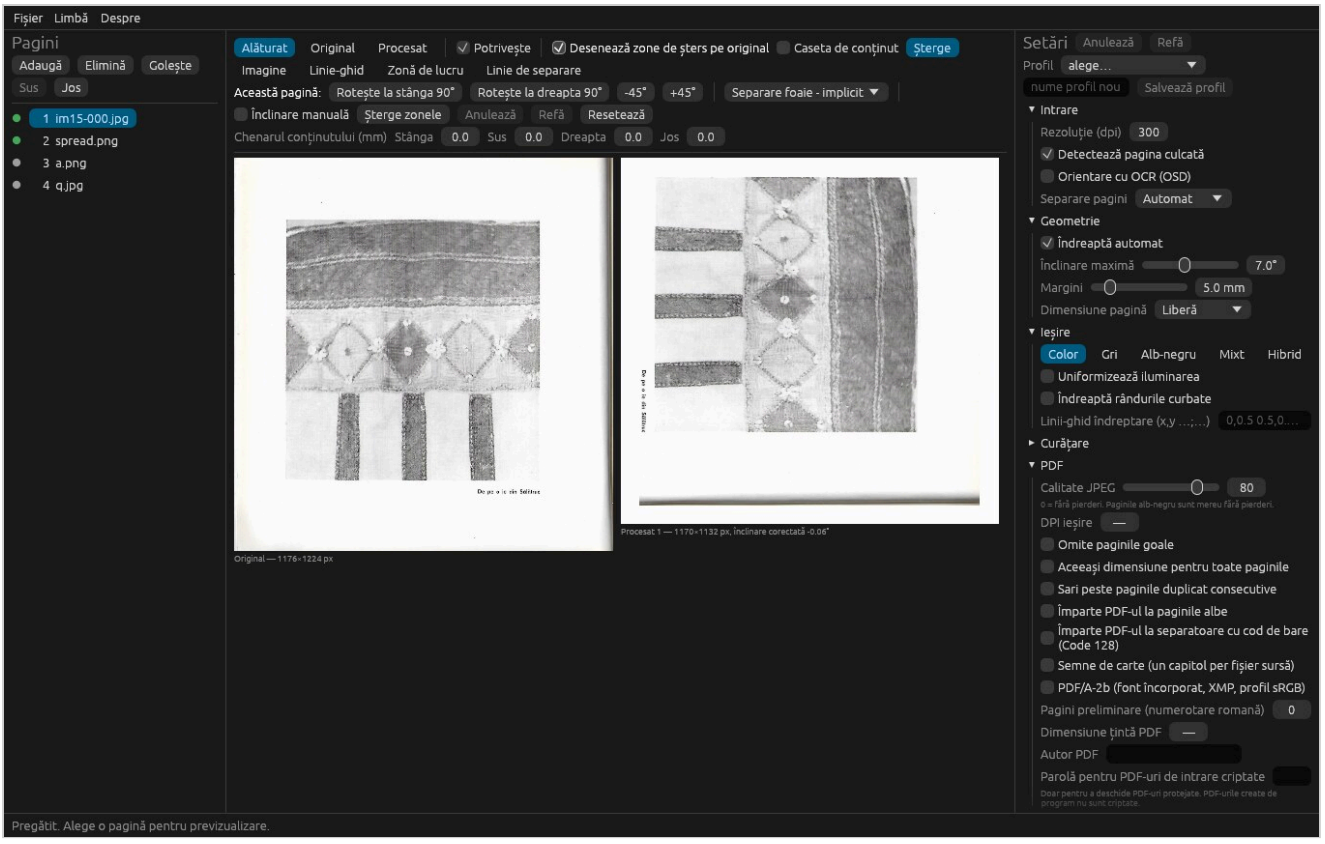

The same page (page 15 of the tester's PDF) and the same settings, apart from the detection: in pre.18, with "Detect page lying on its side" ticked (the default), the program rotated it by 90° although the original is upright (from pre.19–20 it no longer rotates it); with --no-auto-orient (or unticked in the settings) it comes out upright. Checked on the command line; in the window I saw the rotation in pre.18 and its absence in pre.19.



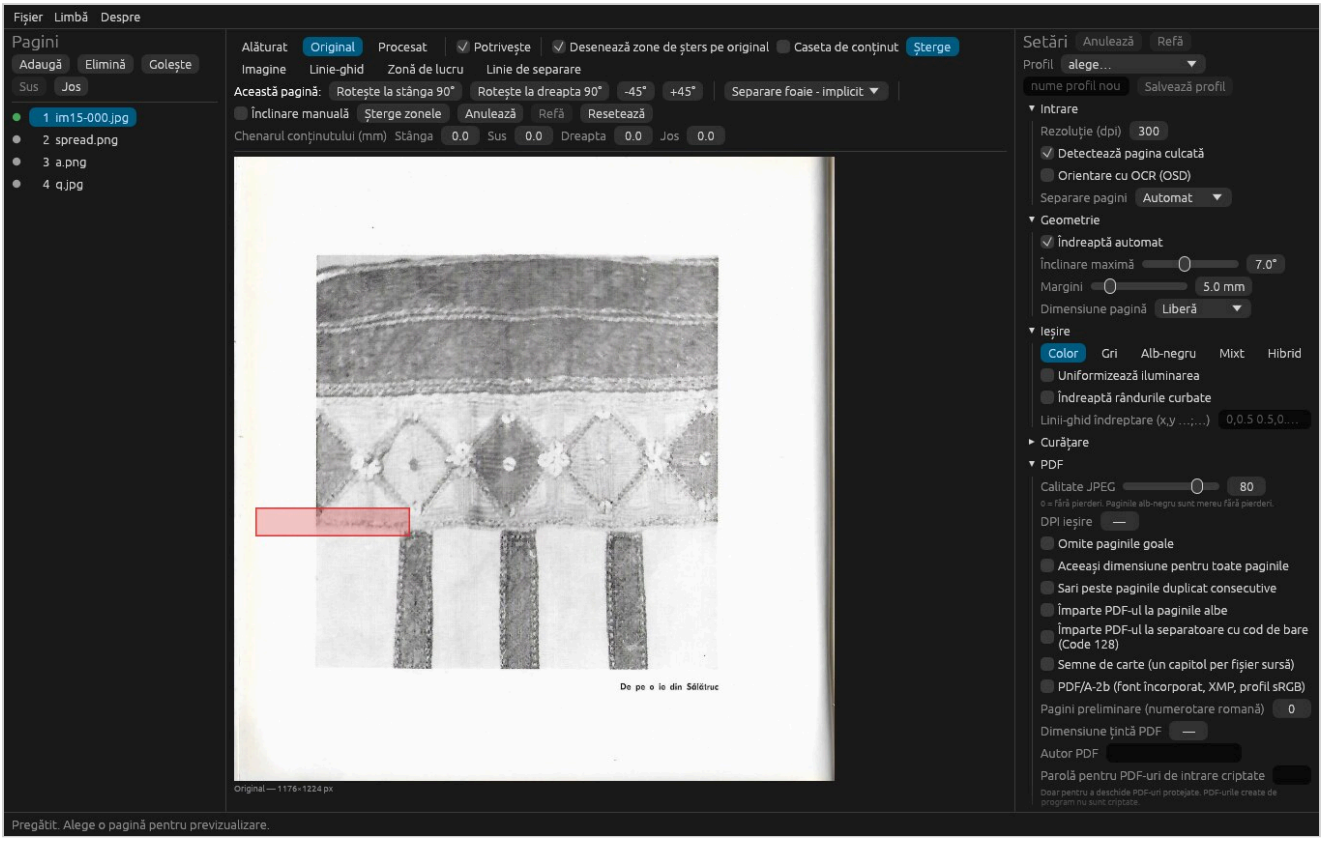

Undo (the button) and Ctrl+Z (with the window in the foreground) remove the last zone drawn. Checked in the window.

## Zones, guides and the content box

Tick "Draw erase zones on the original"; five drawing types appear (Erase, Picture, Guide line, Work area, Cut line):

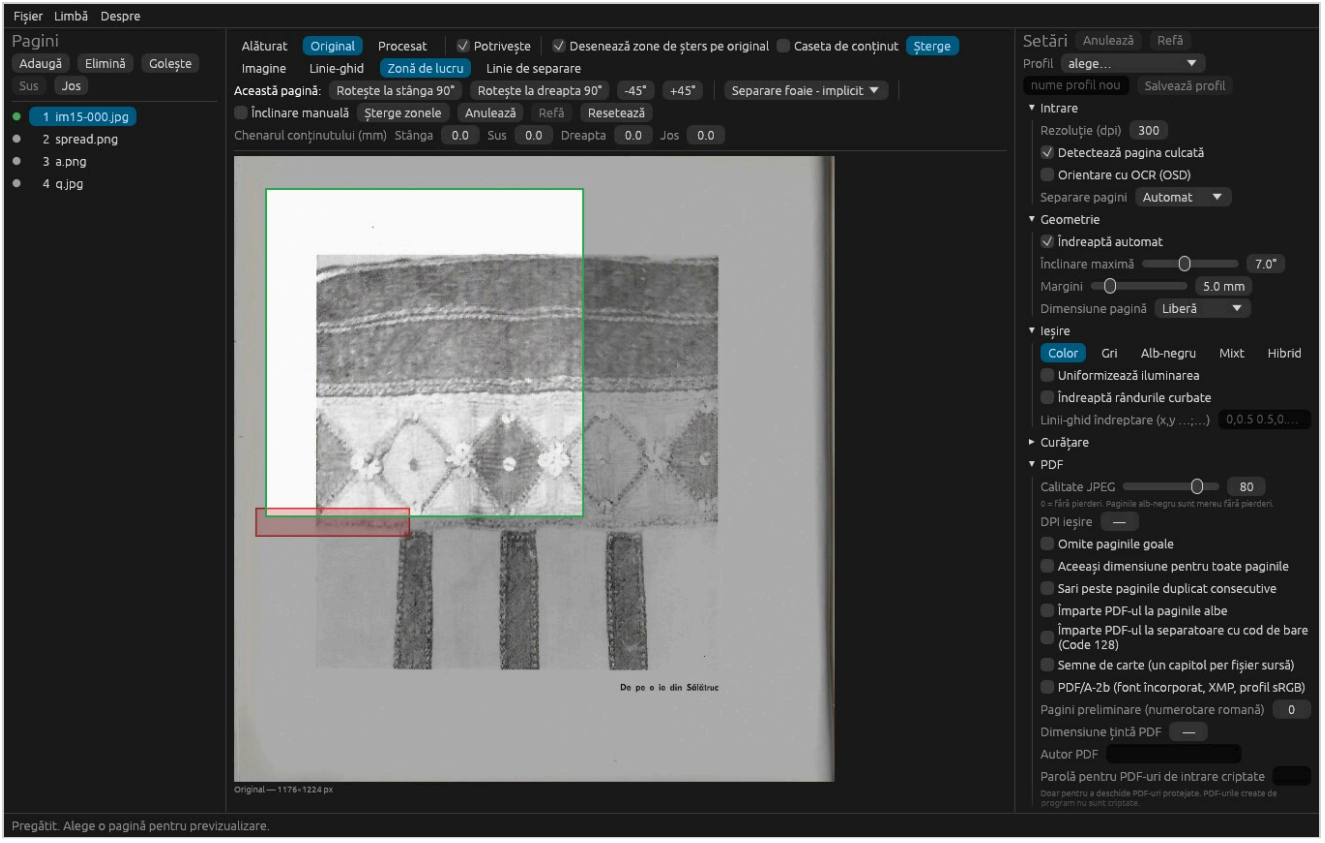



With "Draw erase zones on the original" ticked, the zone types appear: Picture, Guide line, Work area, Cut line (and Erase, the default type).

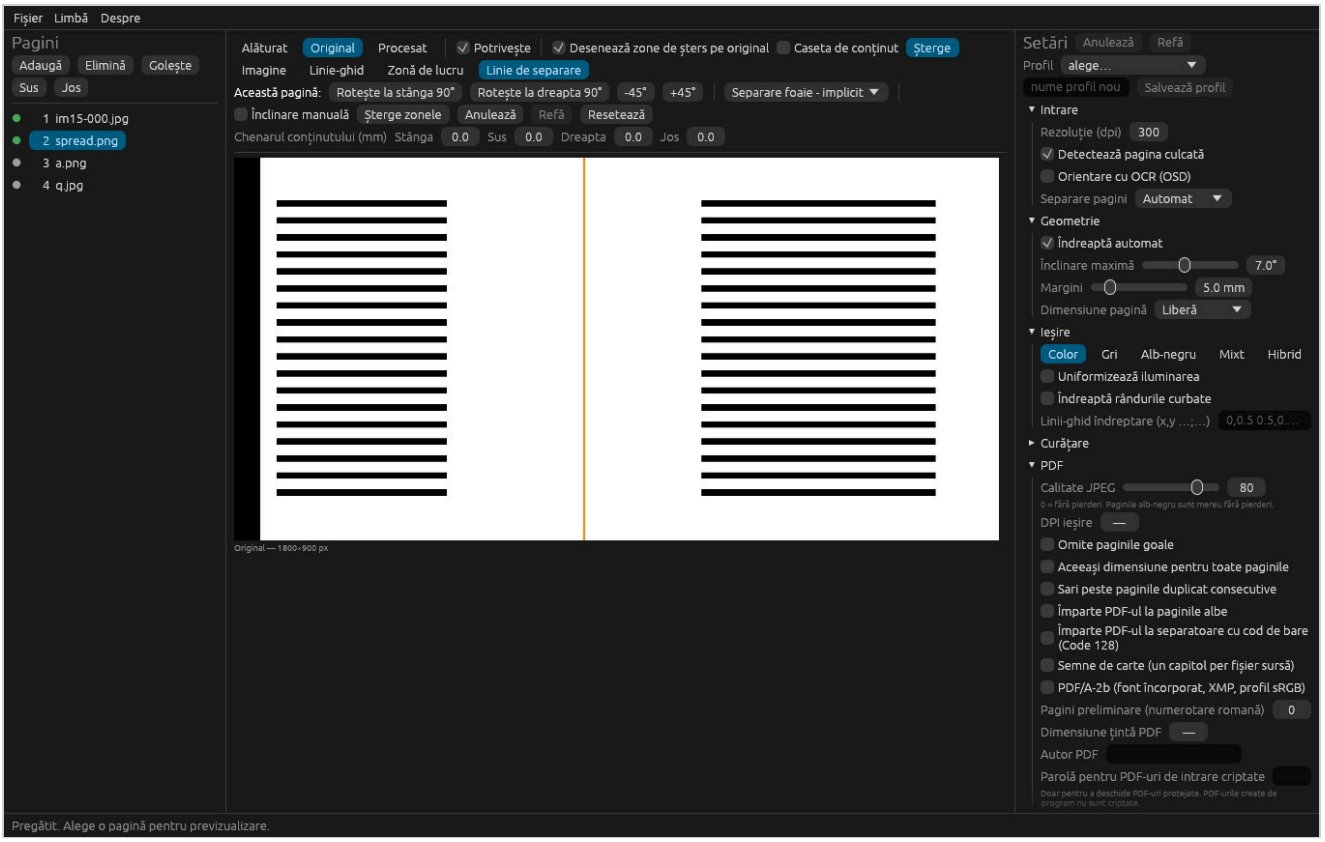



An erase zone (red) drawn with the mouse on the original, in the Original view.

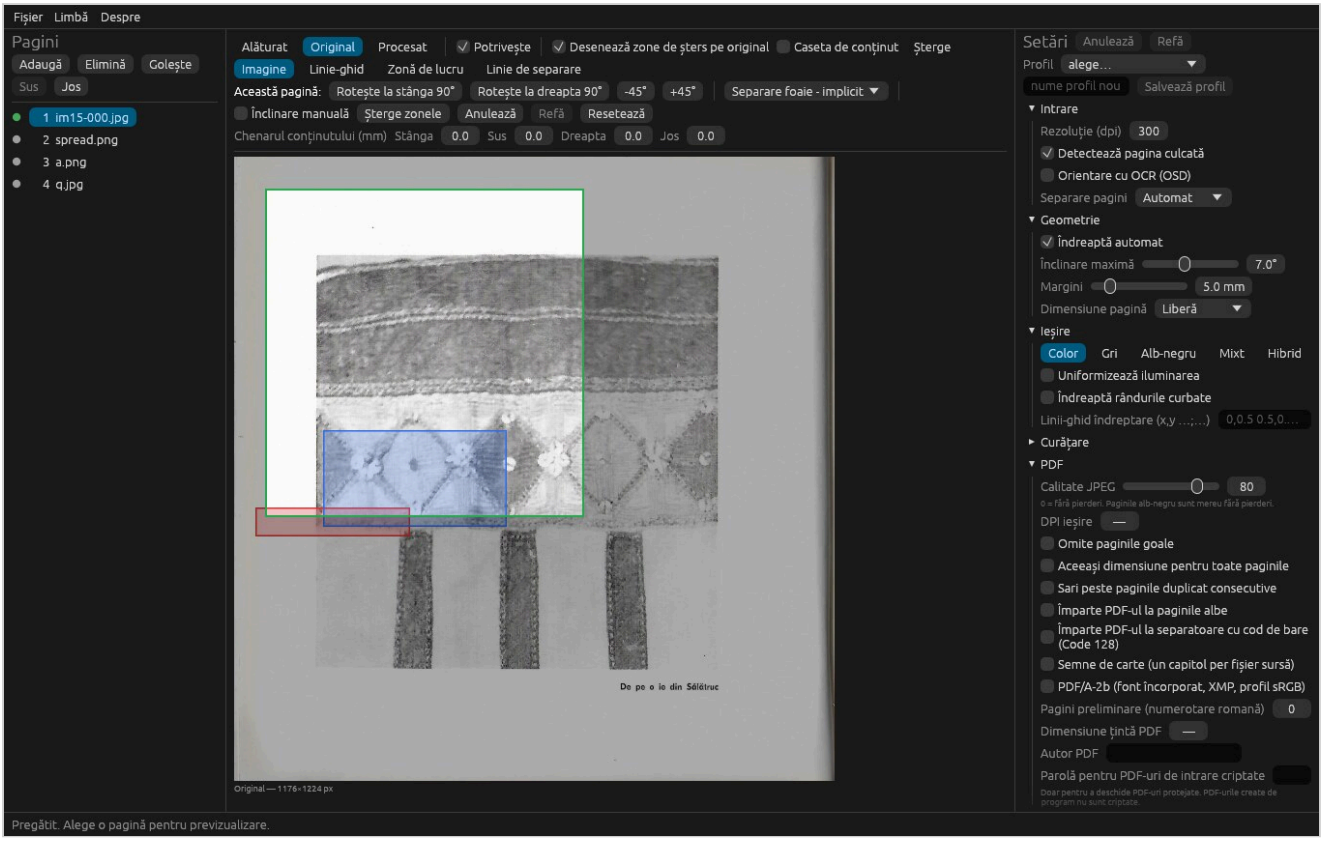
| Type | How you use it | What it is for |
|---|---|---|
| Erase | Drag a rectangle on the original. Right-click on it deletes it. Drag a corner to resize it. | Stain, stamp, someone's note, the edge with a shadow |
| Picture | Same as "Erase". | Tells the program the zone is a picture, not text (it stays in colour in mixed mode) |
| Guide line | Drag freely along a curved line of text. Drag a node to move it; double-click on the line adds a node, double-click on a node deletes it; right-click on the line deletes it. | Straightens the page according to the lines you traced |




The work area (green frame): everything outside it is removed, and the remaining area is seen in the processed page. It also has an erase zone (red).



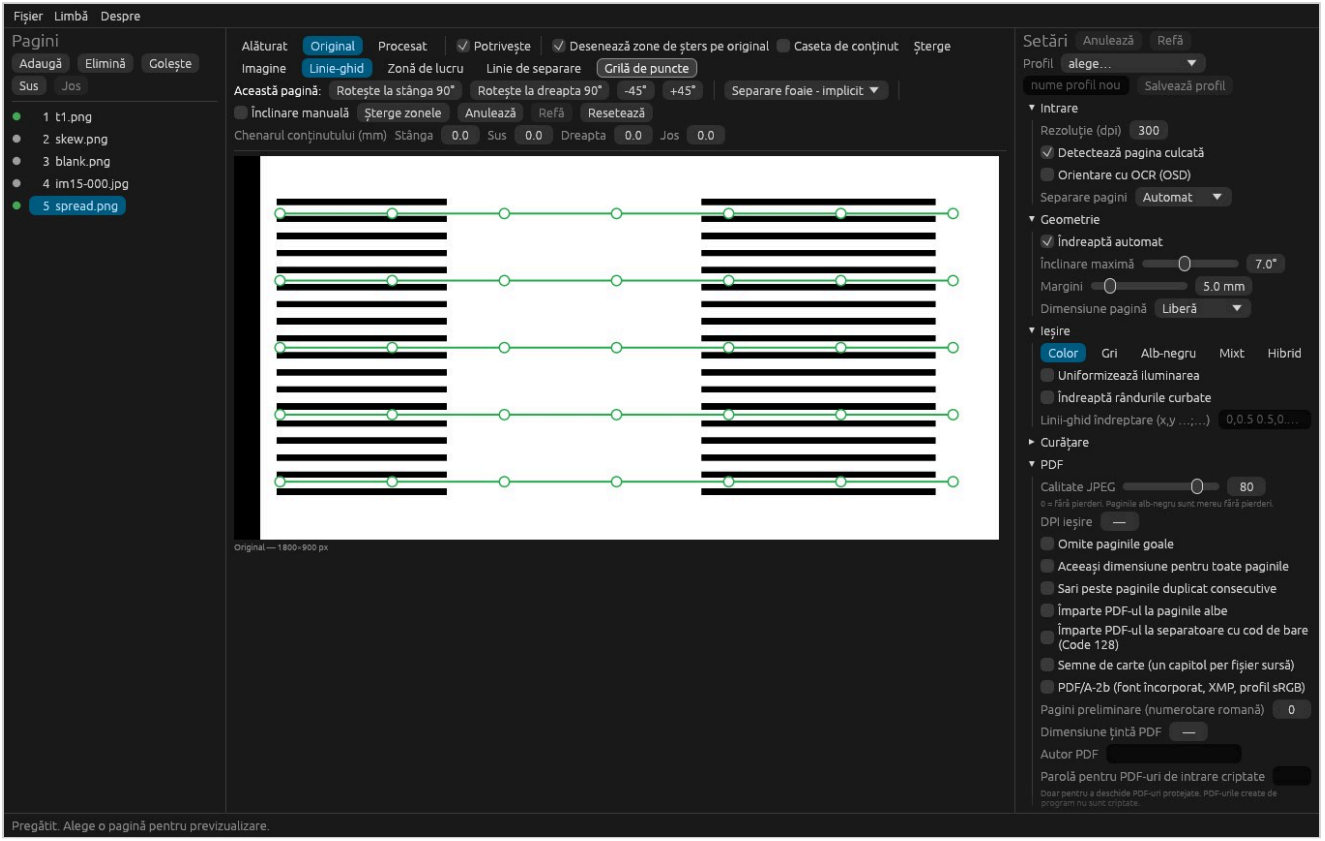

The cut line (orange) moved with the mouse on a double sheet. The example is a synthetic test sheet, with lines instead of text.




A picture zone (blue) drawn on the original. In Colour mode it changes nothing; the effect is seen in Mixed mode (next figure).




The same page and the same picture zone (blue): in Mixed mode the zone stays grey (left), in Black-and-white mode everything becomes black and white (right). The page appears rotated by 90° by the detection of a page lying on its side (see above).




"Grid of points" for Guide line: 5 lines of 7 points each, over the lines of text.

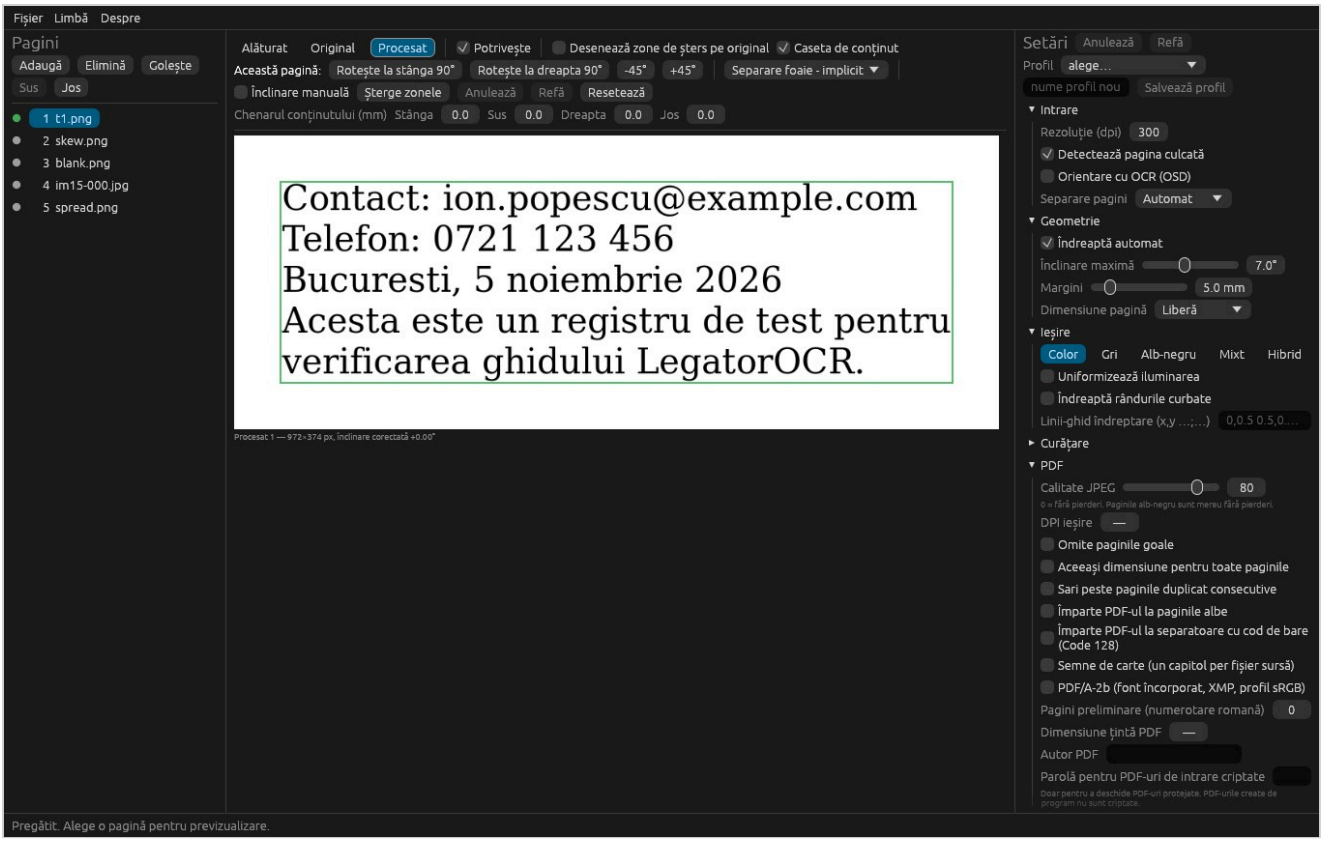
**Grid of points** (for *Guide line*) adds 5 lines of 7 points each, over which you drag the points onto the lines of text. A second click on the button does not add the same grid a second time. If you have already moved points, a new grid is added alongside the moved one.

**Work area** (since edition pre.18): you drag a rectangle on the original; everything outside it is painted with the paper colour before any measurement, so a dark edge or the neighbouring page no longer matters. Right-click on the original removes it. On the command line: `--work-area X,Y,L,H` (millimetres).

**Cut line** (since edition pre.18): by clicking or dragging on the original you move the orange vertical line; the double sheet is cut there, not at the middle or at the gutter. Since pre.21 it

works at any rotation; on sheets rotated by 90° or 270° the line cuts horizontally and the pages come out one below the other (the left page of the original first). Right-click removes it. On the command line: `--cut-at 0.45`.

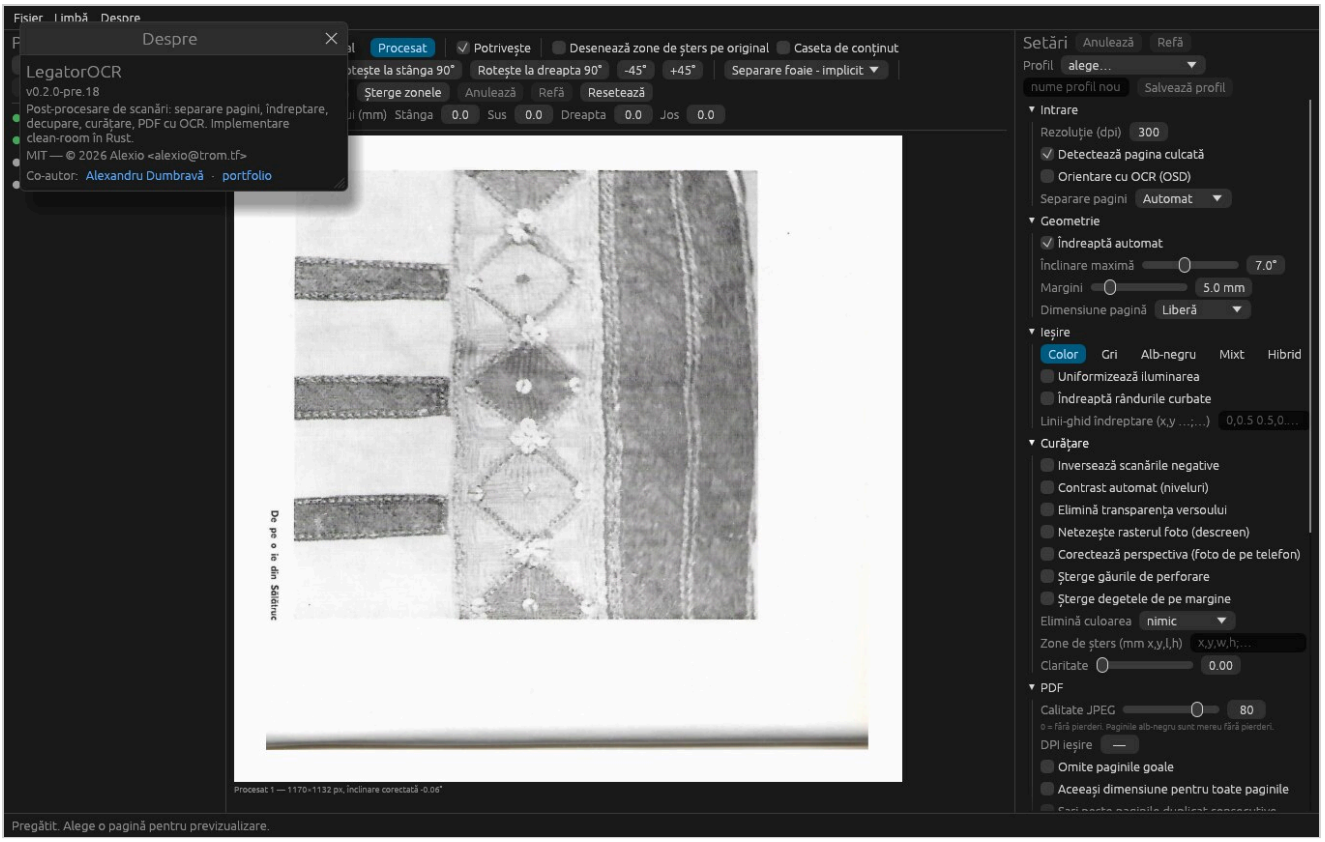
**The content box** (the tick box of the same name) shows a green frame on the processed page. Drag the sides to widen or narrow the kept area. On a double-page sheet, the change applies to both pages.



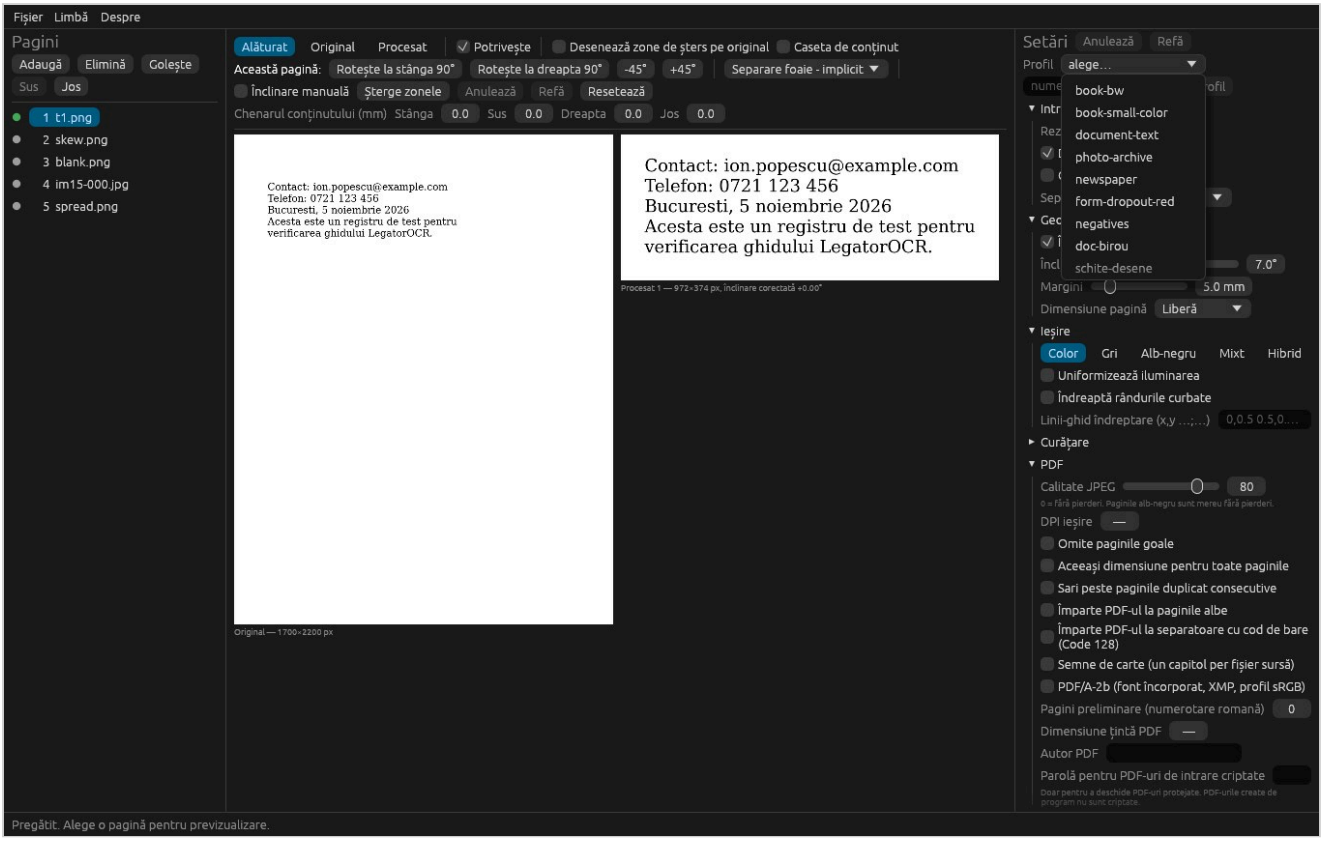

The content box (green frame) on the processed page.

## The settings on the right

They are grouped into sections: Input, Geometry, Output, Cleaning, PDF, OCR, plus the profiles. Section 6 explains what each one does.



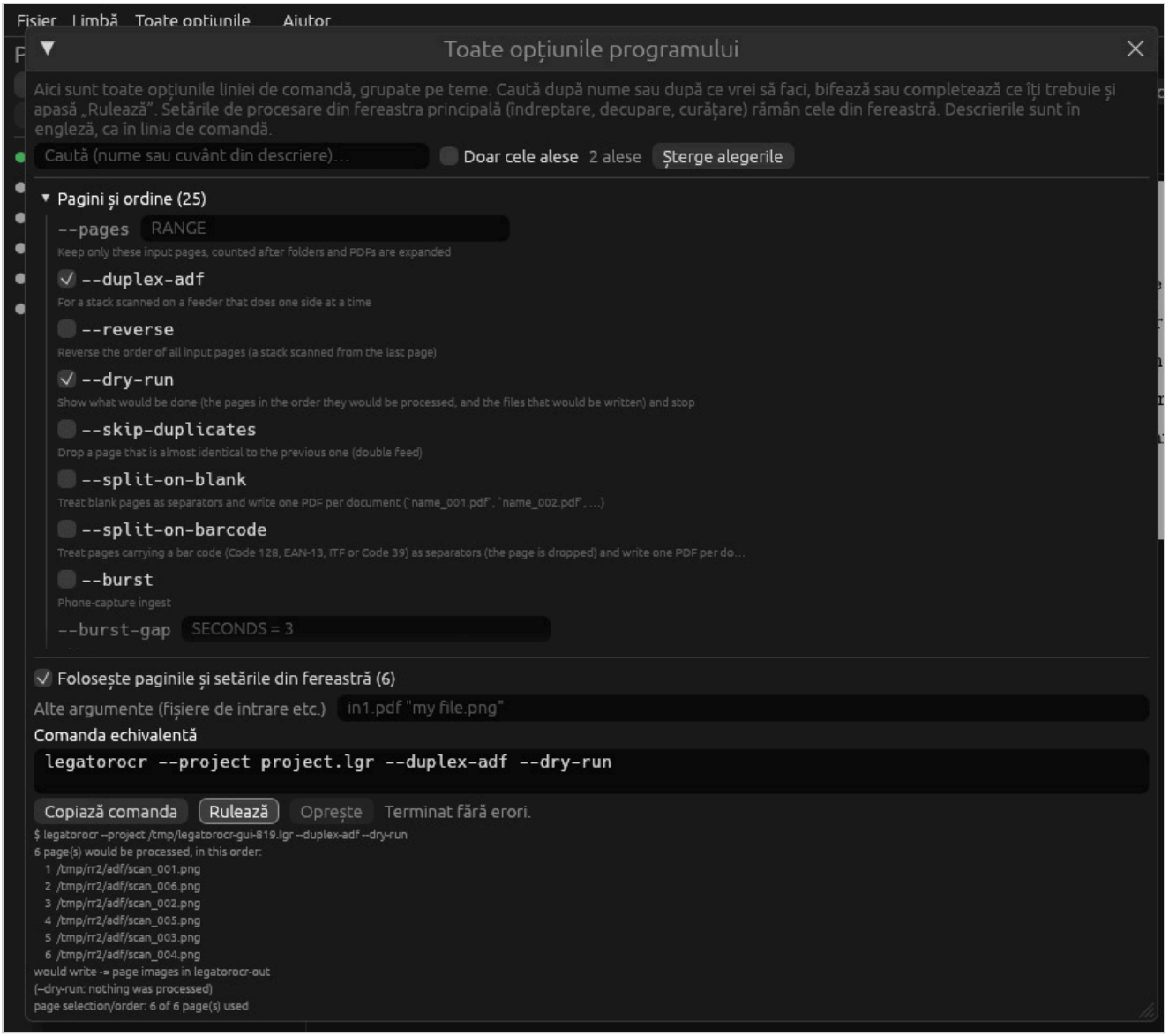

The Cleaning section of the settings, expanded: inversion of negative scans, automatic contrast, perspective, removal of punch holes, erase zones as numbers.




The "Profile" list in the settings (the built-in profiles; yours appear in the same place after "Save profile").

## All options and the user guide

The **All options…** button in the top bar opens a window with **all** the command-line options (over 200), grouped by topic: pages and order; straightening, cropping and cleaning; outputs (PDF, images, documents); OCR and text; archiving, signing and verification; search and comparison; batches, system and modules. The list is read from the command-line program shipped with the window itself, so it always matches the edition you have.



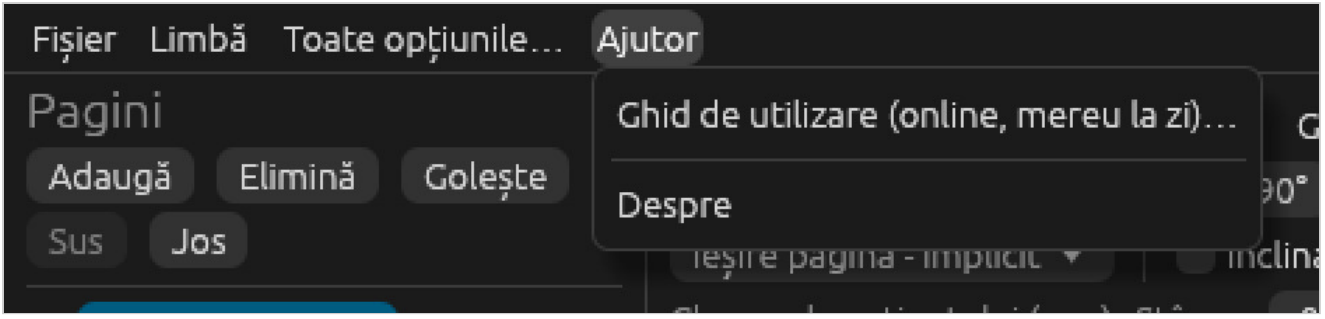

The "All options" window, with the "Pages and order" group open, `--duplex-adf` and `--dry-run` ticked and the result of the run: the order in which the six pages of the stack would be processed (test pages made by me).

- **Search** by name (`pdf`, `dry`) or by a word from the description; the groups with results open by themselves. "Only the chosen ones" shows what you have ticked or filled in.
- Empty boxes are not sent: an option with a value enters the command only when you type something (or choose from a list). The **Choose…** button opens the file or folder dialog. For options that can be repeated (for example `--erase`) you write one value per line.
- **Use the pages and settings from the window** (ticked by default): the command receives the loaded pages, with the adjustments of each, through a temporary project. The

processing settings of the main window (straightening, cropping, cleaning) take priority; here you fill in the outputs and the special functions. Untick it for commands without pages (`--doctor`, `--verify-sums`) or write the input files in "Other arguments".

- **The equivalent command** is updated as you choose; you can copy it and use it in a terminal or in scripts. **Run** executes it and shows the output below; **Stop** interrupts it. Relative paths are relative to your home folder.
- The option descriptions are in English, as on the command line; the rest of the window is in the chosen language.

The **Help** menu has the link to this guide: *User guide (online, always up to date)* opens the latest version published on legatorocr.chiuta.ro, and *Guide of this edition* (it appears when the PDF file is next to the program, as in the published archives) opens the copy shipped with the program. The guide is rebuilt for each edition, and the publication of an edition is stopped if the guide does not mention its version.




Checked: the window, the search, the ticking, the running of a command with the pages from the window (the screenshot above) and the opening of the Help menu, on Linux, on a virtual screen. Not checked: Windows, macOS and arm64; the actual opening of the guide in a browser (I only checked the menu); the "Choose…" dialogs; the translations of the window's new texts (written automatically).

# 5. The command line: recipes for frequent tasks

General syntax: `legatorocr INPUTS... [options]`. The inputs can be images (PNG, JPEG, TIFF, BMP), scanned PDFs or a folder. `legatorocr --help` lists all the options; `--list-profiles` shows the profiles. The options in the recipes are checked automatically before each edition, so they really exist; the recipes themselves have not all been tried on real material.

### Scanned book → PDF with searchable text

```
legatorocr scans/*.png --profile book-bw --ocr --lang ron --pdf carte.pdf --pdfa --txt
carte.txt -o pages
```

`--pdfa` writes PDF/A-2b for archiving, and `--txt` also saves the text. `-o pages` is the folder for the processed images.

```
$ legatorocr scans/*.png --profile book-bw --ocr --lang ron --pdf carte.pdf --pdfa --txt carte.txt -o pagesT
OCR engine: tesseract 5.3.4, language: ron
a_L: 304x164 px @ 300 dpi, skew +0.00°
a_R: 284x254 px @ 300 dpi, skew +0.00°
p15b: 1162x1232 px @ 300 dpi, skew +0.00°
spread_L: 524x818 px @ 300 dpi, skew +0.00°
spread_R: 674x818 px @ 300 dpi, skew +0.00°
text: 972x374 px @ 300 dpi, skew +0.00°
wrote 6 page image(s) to pagesT
wrote carte.pdf (6 page(s), 27 OCR words)
wrote carte.txt
```

The recipe on 4 test files: 6 pages in carte.pdf. With --pdfa the program writes PDF/A-2b.

## Phone photos → clean pages

```
legatorocr photos/*.jpg --profile phone-photos --pdf pagini.pdf -o pages
```

The profile evens out the lighting, corrects perspective, wipes the background around the page (`--trim-background`), cuts open books in the middle, widens the skew range and removes fingers without touching the text. A page that comes out without text is read a second time on its content only. For uniform A4 pages, with enlarged text, use `--profile phone-photos-a4`. Several photos of the same page can first be combined with `--fuse-frames` (it helps with strong noise; with moderate noise it does not). The option takes an output file: legatorocr photo1.jpg photo2.jpg photo3.jpg --fuse-frames page.png, then you process page.png as usual; the first photo is the reference.

```
$ legatorocr f1.jpg f2.jpg f3.jpg --fuse-frames fused.png
f1.jpg: aligned (shift 0,0)
f2.jpg: aligned (shift -3,-2)
f3.jpg: aligned (shift 2,-3)
wrote fused.png from 3 of 3 frames
```

--fuse-frames with three photos of the same page: it aligns them and writes fused.png from 3 of 3 frames.

## Archive copy with proof of integrity

```
legatorocr scans/*.png --profile registre-manuscrise --ocr --lang ron --pdf registru.pdf --
pdfa --checksums --archive-package registru-arhiva -o pages
legatorocr --verify-sums registru-arhiva/SHA256SUMS.txt
```

The package keeps the originals, the project, the OCR words with confidence, the outputs and a `RUN.txt` that says how to resume the work. For copies on other disks: `--backup-from` and `--backup-to`.

```
$ rm -rf registru-arhiva2; legatorocr scans/text.png --profile registre-manuscrise --ocr --lang ron --pdf registru2.pdf
--pdfa --checksums --archive-package registru-arhiva2 -o pagesT2; legatorocr --verify-sums registru-arhiva2/SHA256SUMS.t
xt
OCR engine: tesseract 5.3.4, language: ron
text: 972x374 px @ 300 dpi, skew +0.00°
wrote 1 page image(s) to pagesT2
wrote registru2.pdf (1 page(s), 20 OCR words)
wrote registru-arhiva2/derived/ocr.json
archive package: registru-arhiva2 (6 files, 1 originals)
wrote ./SHA256SUMS.txt (2 file(s) added or updated)
5 ok, 0 mismatched, 0 missing
```

The archive package created and verified: "5 ok, 0 mismatched, 0 missing".

## Hiding names and numbers before sharing

```
legatorocr scan.png --ocr --lang ron --redact email,phone-ro --format none --pdf redactat.pdf
```

The words found are blacked out in the image and removed from the text. **Check the result with your own eyes:** a word that OCR did not recognise is not hidden.

```
$ legatorocr scans/text.png --ocr --lang ron --redact email,phone-ro --format none --pdf redactat.pdf; pdftotext redacta
t.pdf - | head -4
OCR engine: tesseract 5.3.4, language: ron
text: 972x374 px @ 300 dpi, skew +0.00°
redaction: preset email: 0 covered; preset phone-ro: 1 covered;
warning: preset email matched nothing in 1 page(s). OCR may have misread the value (an "@" is often read as "G" or "(a")
: look at the pages yourself before you hand the files over.
warning: 1 word(s) left in the text still look like personal data (page 1: address-like word). They were NOT covered; ch
eck them by eye.
wrote redactat.pdf (1 page(s), 17 OCR words)
Contact: ion.popescuGexample.com

Telefon:
Bucuresti, 5 noiembrie 2026
```

Real result: the phone number was hidden, but the e-mail address was NOT, because OCR read "@" as "G" (ion.popescuGexample.com) and the e-mail pattern did not match. This is why you must check the result with your own eyes. Since pre.20 the program writes at the end how many words each pattern covered and warns when a pattern found nothing or when words that look like addresses or long numbers remain in the text, exactly as in the screenshot above.

## A large batch on several computers

```
legatorocr scans/*.png --shard 1/3 --profile book-bw --pdf part1.pdf -o pages1
legatorocr --merge-pdf part1.pdf part2.pdf part3.pdf --pdf carte.pdf
```

Run the same list on each machine, with a different `--shard`. For low memory: `--max-memory 500`. For unknown or hostile files: `--isolate-decoders`.

```
$ legatorocr scans/*.png --shard 1/2 --profile book-bw --pdf part1.pdf -o pages1; legatorocr scans/*.png --shard 2/2 --p
rofile book-bw --pdf part2.pdf -o pages2; legatorocr --merge-pdf part1.pdf part2.pdf --pdf carte-m.pdf
shard 1/2: pages 1..2 of 4
a_L: 304x164 px @ 300 dpi, skew +0.00°
a_R: 284x254 px @ 300 dpi, skew +0.00°
p15b: 1162x1232 px @ 300 dpi, skew +0.00°
wrote 3 page image(s) to pages1
wrote part1.pdf (3 page(s))
shard 2/2: pages 3..4 of 4
spread_L: 524x818 px @ 300 dpi, skew +0.00°
spread_R: 674x818 px @ 300 dpi, skew +0.00°
text: 972x374 px @ 300 dpi, skew +0.00°
wrote 3 page image(s) to pages2
wrote part2.pdf (3 page(s))
wrote carte-m.pdf
```

The batch split in two (--shard 1/2 and 2/2) and then joined with --merge-pdf.

## Signing a PDF

```
LEGATOROCR_P12_PASS=... legatorocr --sign-pdf registru.pdf --sign-p12 certificat.p12 --sign-
name "Name"
legatorocr --verify-pdf-signatures registru.signed.pdf --sign-ca ca.pem
```

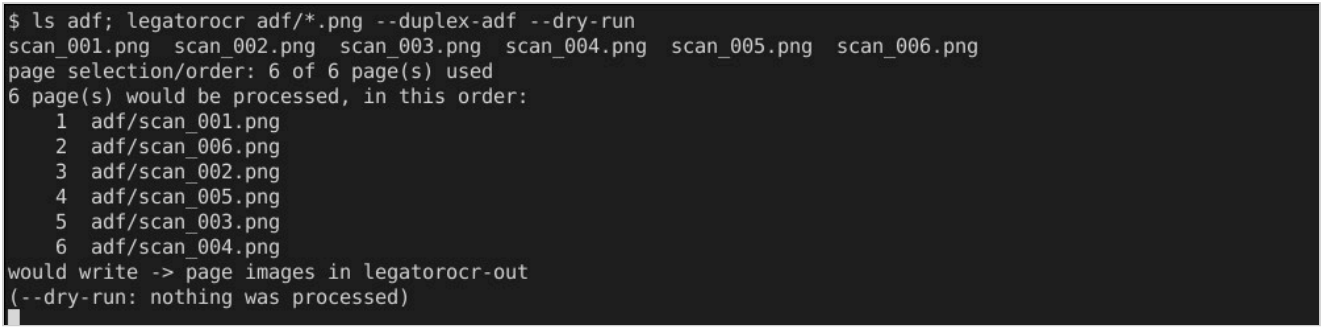
The legal level of the signature is that of your certificate. The certificate password is given through the environment variable, not on the command line. The time stamp (`--sign-timestamp`) was checked only with a local authority over HTTP, not with a real one over HTTPS.




Signing and verification with a test certificate created by me (not a real certificate). "time stamp inside: none": I did not request a time stamp.

## A stack scanned on a feeder, page order and range

Many scanners with a feeder pass the stack once for each side: first all the fronts (1, 3, 5…), then all the backs, in reverse order (6, 4, 2). The files come out in scanning order, not in the order of the book. `--duplex-adf` puts them in reading order: front 1, back 1, front 2, back 2… The number of pages must be even (otherwise the program stops with a message).




Six files in scanning order (scan_001 … scan_006). `--dry-run` shows the order in which they would be processed and which files would be written, then stops without processing anything: scan_006 ends up in position 2, scan_005 in position 4.




The same command without `--dry-run`, with native OCR: the text comes out in the order "Pagina 1 … Pagina 6" (test pages made by me, not a real book).

Other ordering and selection options (they can be combined; the range applies to the numbering of the input files, starting from 1):

- `--pages 1-10,15,20-`: keeps only the pages indicated; `20-` means "from 20 to the end". It also works on pages taken from PDFs or from multi-page TIFF sheets.

- `--reverse`: reverses the order (a stack scanned from the last page).
- `--dry-run`: checks a large batch before the long run.

Checked: the order and the range with automated tests and with six test pages; I did not try a real scanner with a feeder. The options can also be chosen from the window, under "All options…" (section 4).

## Clean text: your own corrections, running heads and cedilla

For the `.txt` and `.docx` files (the text layer in the PDF, hOCR and JSON keep the words as read, unchanged):

- `--clean-text`: ș and ț with a comma below, words broken at the end of a line joined (with the hyphen kept when the two halves are words from the list and simple joining is not), typographic quotation marks, spaces next to punctuation. `--cedilla` writes ş and ţ instead, for programs and fonts that do not know the comma letters.
- `--find-replace FILE`: your own corrections, one per line, `wrong => right`. Lines that start with `#` are comments. Only whole words match (`Ion => Ioan` does not touch "Ionescu") and they are applied in the order of the file, after `--clean-text`. Useful for proper names, abbreviations or a stamp that the engine always misreads in the same way.
- `--drop-running-heads`: leaves out the printed running heads, footers and page numbers: the first or last line of a page that repeats (digits do not count) on at least half of the pages; at least four pages are needed. The lines removed are listed on stderr, so that you can see what went.

```
$ cat reguli.txt
O="--engine native --ocr --lang ron"
legatorocr rev/p01.png $O --txt f0.txt 2>/dev/null
legatorocr rev/p01.png $O --txt f1.txt --find-replace reguli.txt 2>/dev/null
echo "fără regulă:"; grep -h "secretarul\|ha\." f0.txt
echo "cu regulă:"; grep -h "secretarul\|ha\.\|hectare" f1.txt
# corecturi proprii
Ionescu => Ionescu-Vale
ha => hectare
Volumului => Volumul
fără regulă:
în anul 1951 de către secretarul Ionescu.
și suprafața înscrisă în ha. Corecturile se
cu regulă:
în anul 1951 de către secretarul Ionescu-Vale.
și suprafața înscrisă în hectare. Corecturile se
```

The rules file (first block) and the same two lines read from the test page without and with `--find-replace`: "Ionescu" became "Ionescu-Vale" and "ha" became "hectare".

```
$ legatorocr rev/*.png --engine native --ocr --lang ron --clean-text --drop-running-heads --txt rev.txt; head -
OCR engine: LegatorOCR native engine 0.2.0-pre.26, language: ron
p02: 996x904 px @ 300 dpi, skew +0.00°
p01: 992x904 px @ 300 dpi, skew +0.00°
p04: 998x904 px @ 300 dpi, skew +0.00°
p03: 998x904 px @ 300 dpi, skew +0.00°
p05: 998x904 px @ 300 dpi, skew +0.00°
p06: 998x904 px @ 300 dpi, skew +0.00°
wrote 6 page image(s) to legatorocr-out
6 running head/foot/page-number lines left out of the text (e.g. "Anuarul Văii Mari 1", "Anuarul Văii Mari 2",
Văii Mari 3")
wrote rev.txt
Pagina 1
Registrul comunei Valea Mare a fost rescris
în anul 1951 de către secretarul Ionescu.
Fiecare pagină păstrează numele proprietarilor
```

On six test pages with the running head "Anuarul Văii Mari" and a number, the program removed six running-head lines; the text starts directly with "Pagina 1". Note: the rule removes any first/last line that repeats, so in a book where the last line of each page is similar (for example "Shelf position: 14") those lines would go too; this is why they are listed. The function was checked with tests and on a few pages, not on a large set of books.

### Other recipes

The complete recipes are in `docs/tasks/` in the repository, in English and Romanian (document verification with `--verify-document` and `--mrz`, archive copy, large batch).

## 6. Important settings and when to use them

### Profiles

```
$ legatorocr --list-profiles | cut -c1-100
book-bw              Text books: auto threshold, black & white, punch holes removed
book-small-color     Colour books: sharp text over a light background, smallest files
document-text        Office documents: sharp text, pictures kept in colour
photo-archive        Archive quality colour, light sharpening
newspaper            Newsprint: Wolf threshold, stronger speck removal, flattened curl
form-dropout-red     Forms with red rules or stamps: drop red, erase ruled lines
negatives            Negative scans: invert and stretch contrast
doc-birou            Office documents and invoices (scan at 300 dpi): black & white, searchable PDF
schite-desene        Architectural / pencil sketches (scan at 400 dpi): grey, even lighting, light c
registre-manuscrise  Old registers and manuscripts (scan at 300-600 dpi): colour kept faithfully, hi
flux-1-optim         Flux 1 (optim, scan at 300 dpi): sharp 1-bit text, pictures in colour, margins
flux-2-draft         Flux 2 (ciornă / draft): quick grey pass, light compression - for checking and
flux-3-volum         Flux 3 (volum mare / high volume): 1-bit text over a very light background, sma
diploma              Diplomas and certificates (scan at 300-400 dpi): colour seals, stamps and signa
registru-stare-civila Civil-register pages with handwriting and stamps (scan at 400 dpi): colour kep
harta                Maps (scan at 400 dpi or more): colour kept faithfully, no thresholding, no des
partitura            Music scores (scan at 600 dpi): fine 1-bit with a local threshold so thin staff
phone-photos         Pages photographed by phone: level lighting, wider skew range, fingers removed
smokeprof            user profile
```

legatorocr --list-profiles. The last line (smokeprof) is a test profile of mine, not part of the program.

A profile is a ready-made set of settings for a type of material. It is chosen from the "Profile" section (interface) or with `--profile NAME` (command line). You can have your own profiles: you configure the settings, type a name and press "Save profile".

| Profile | What for |
|---|---|
| `book-bw` | books with text, black and white, punch holes removed |
| `book-small-color` | colour books, clear text on a light background, small files |
| `document-text` | office documents, images kept in colour |
| `photo-archive` | photographs, archival quality |
| `newspaper` | newspapers: Wolf threshold, stains removed more strongly |
| `form-dropout-red` | forms with red lines or stamps |
| `negatives` | negative scans |
| `phone-photos` | phone photos: perspective corrected, background around the page wiped, open books cut in the middle, fingers removed |
| `phone-photos-a4` | like `phone-photos`, but each page on an A4 sheet, with the text enlarged to fill the sheet (also pages with few lines) |
| `registre-manuscrise` | old registers and manuscripts (300–600 dpi), colours kept |
| `schite-desene` | sketches and pencil drawings (400 dpi) |
| `doc-birou`, `flux-1-optim`, `flux-2-draft`, `flux-3-volum` | flows for office documents, from maximum quality to high volume |
| `diploma`, `registru-stare-civila`, `harta`, `partitura` | **not verified on real samples** |

The complete list is given by `--list-profiles`.

## The settings that matter most

| Setting | What it does | When you change it |
|---|---|---|
| Resolution (dpi) | what resolution the pages were scanned at | when the image has no dpi in its header; for text 300 |
| Auto-deskew / maximum skew | the limit of the automatically corrected angle | when a page that comes out "borderline" is in fact crooked |
| Margins, page size, same size | margins in mm; free page size, custom or identical for the whole book | for books with mixed formats |
| Output mode | colour, grey, black and white, mixed (1-bit text + colour images), hybrid (+ low-resolution background) | black and white for pure text; mixed for text with pictures |
| Threshold method | Sauvola (default), Wolf, Niblack, NICK, Bradley, Otsu or automatic | when the text comes out faded or full of stains; `auto` tries several on each page |
| Sensitivity | the sensitivity of the Sauvola method for black-and-white output | default 0.3; change it in small steps and compare the result on the same page |
| Picture sensitivity | how easily an area is taken to be a picture | in mixed/hybrid mode |
| Remove small specks | the strength of speck cleaning (0.25–4; 1 = default) | lower it if dots or diacritics disappear; raise it if specks remain |
| Dewarp (curved lines) | straightens the arched lines near the spine | for thick books; skipped if no clear curvature is seen |
| Even out lighting | evens out the background | for photos with shadow or uneven lighting |
| JPEG quality | 0 = lossless; black-and-white pages are always lossless | for smaller files (suggestion: try 80–90 and compare size and appearance) |
| Output DPI | shrinks the output image to this dpi ("—" = do not shrink) | for smaller PDFs; pages under 250 dpi are automatically enlarged before OCR, so too big a reduction does not help recognition |
| PDF/A-2b | PDF for archiving, with font and colour profile included | for deposit in archives; it has not been checked with veraPDF on the program's outputs |

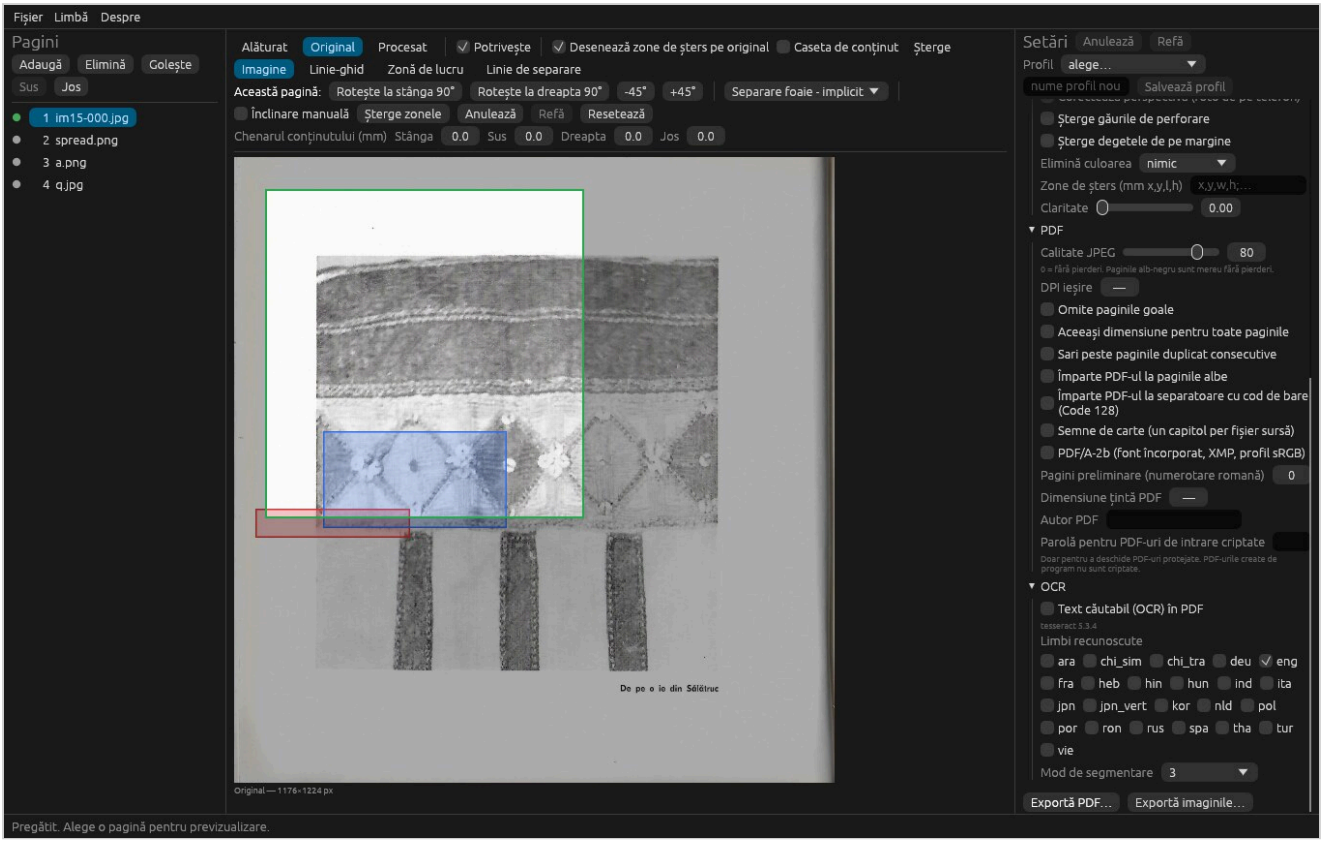
### The formats of the text fields

These fields exist in the settings, but it is easier to draw on the original (section 4):

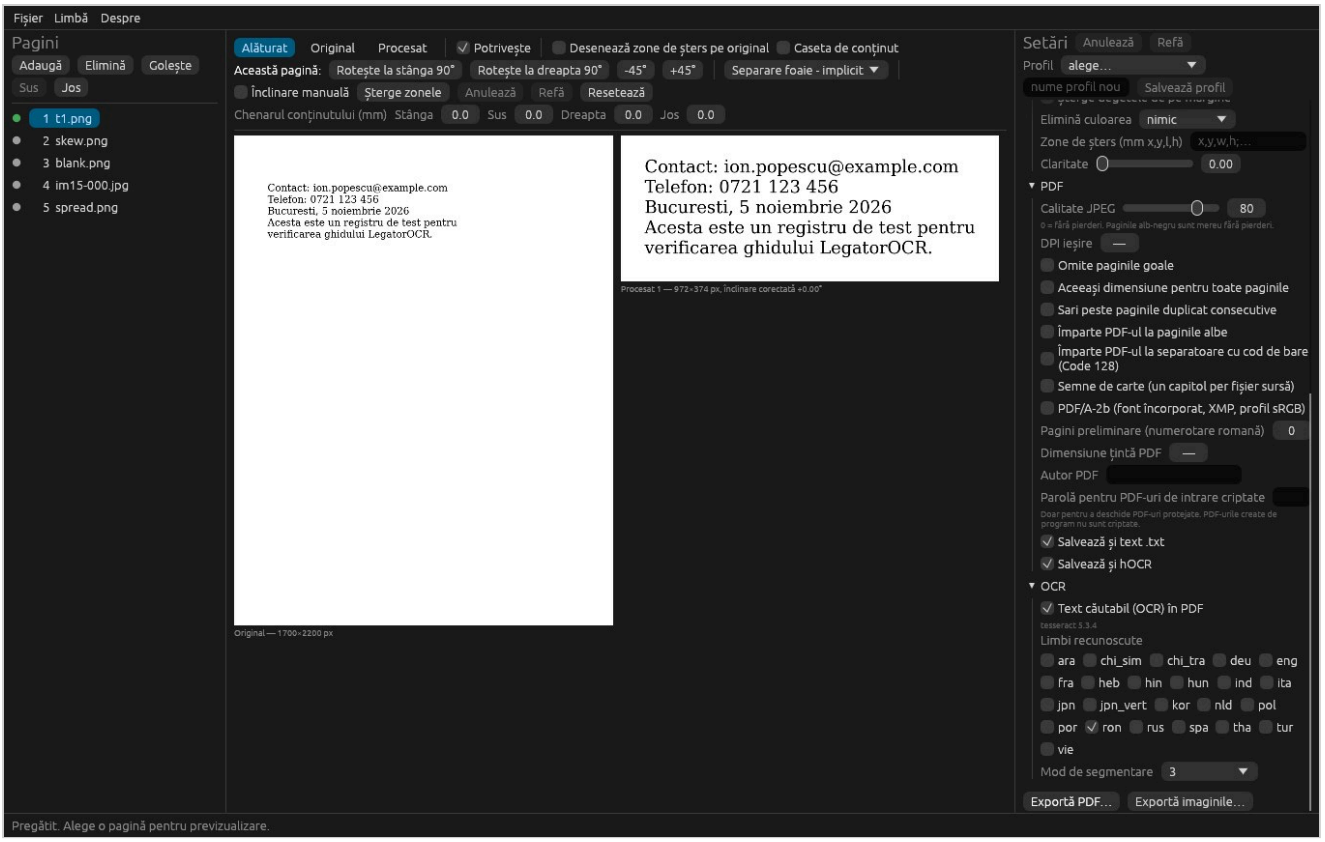
- **Erase zones** and **Picture zones**: `x,y,w,h` in **millimetres** from the top-left corner of the page. Several zones are separated with semicolons: 10,20,30,40;50,60,20,20. At most 64; width and height must be positive. If the form is wrong, the field turns red.
- **Straightening guide lines**: `x,y x,y x,y;x,y x,y x,y`, with **fractions between 0 and 1** of the page already straightened (0,0 = top-left corner). Each line has at least two points; at most 64 lines in total.
- On the command line: `--erase X,Y,W,H` (repeatable) and `--dewarp-guide "X,Y X,Y …"` (repeatable).

## 7. OCR: engines, languages and accuracy

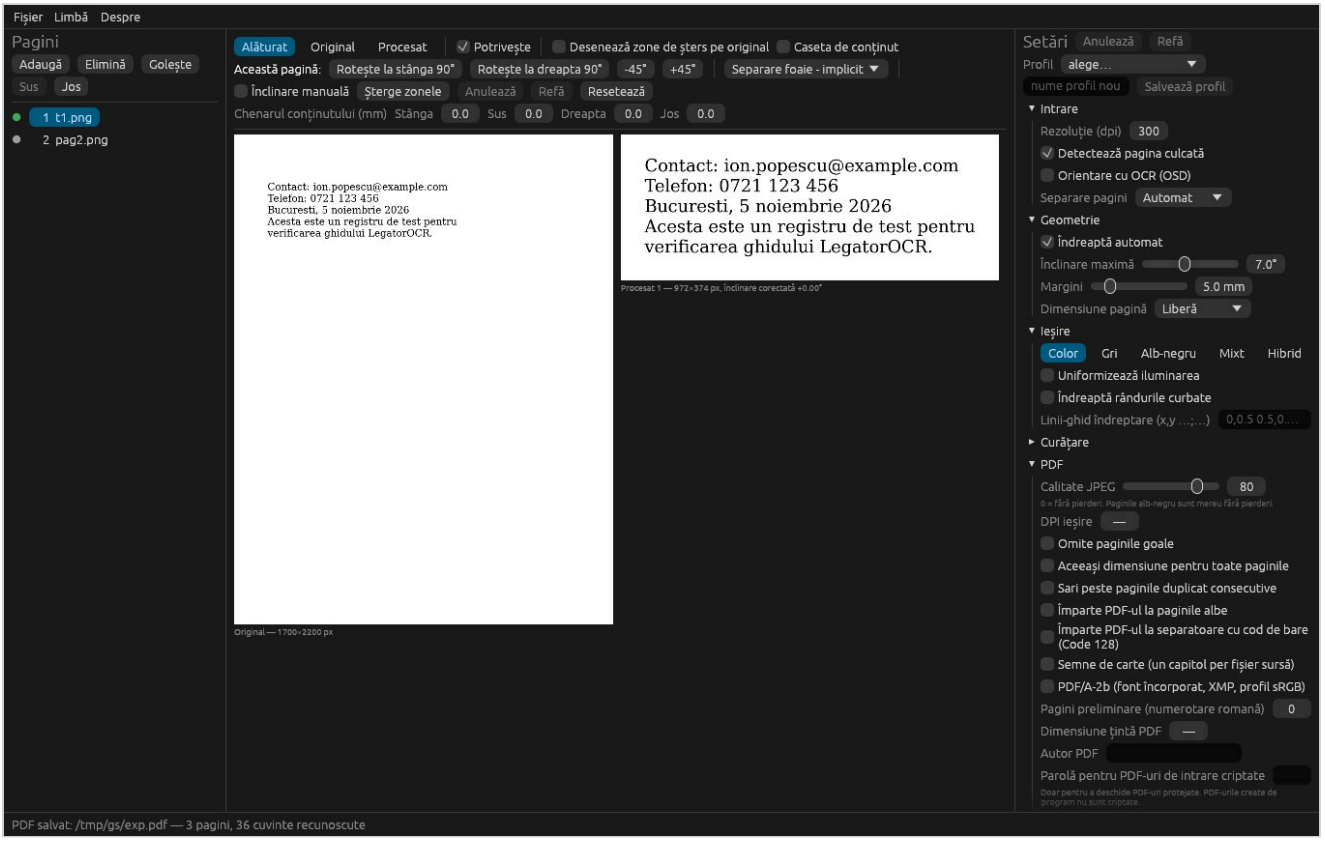
The "Searchable text (OCR) in PDF" tick box (or `--ocr` on the command line) adds an invisible text layer over the image. The language is chosen from "Recognised languages" (or `--lang ron`); it can be combined, for example `ron+eng`. The languages of the interface and of the built-in engine: Romanian, English, French, Italian, Spanish, Portuguese, German, Hungarian.




The OCR section of the settings: the "Searchable text (OCR) in PDF" tick box, the recognised languages (those installed in Tesseract) and the segmentation mode. In the screenshot Tesseract 5.3.4 was installed, which is why all its languages appear; I used the ron choice in the next figure.




With OCR ticked, "Also save .txt text" and "Also save hOCR" appear. Here the chosen language is ron.




After the export, the bottom bar shows: "PDF saved: … — 3 pages, 36 words recognised". (I started the export with a test mechanism of the program, not from the file dialog.)

## Three engines

| Engine | How you choose it | Notes |
|---|---|---|
| Native (the program's own) | `--engine native` | no installation, written in Rust; model trained on synthetic text, then adjusted on lines from a Romanian magazine and from a 1960 textbook |
| Tesseract | `--engine tesseract` | separate program, see section 2; used automatically when installed and has the requested language (`--engine auto`, the default) |
| ONNX bridge | `--engine tesseract --tesseract scripts/engine-onnx.py` | for advanced users; requires Python, onnxruntime and separately downloaded models |

```
$ legatorocr scans/text.png --engine native --ocr --lang ron --txt nat.txt --format none; cat nat.txt; echo; legatorocr
scans/text.png --engine tesseract --ocr --lang ron --txt tes.txt --format none; cat tes.txt
OCR engine: LegatorOCR native engine 0.2.0-pre.19, language: ron
text: 972x374 px @ 300 dpi, skew +0.00°
wrote nat.txt
Contact: ion.popescu(aexample.com
Telefon: 0721 123 456
Bucuresti, 5 noiembrie 2026
Acesta este un registru de test pentru
verificarea ghidului LegatorOCR.
OCR engine: tesseract 5.3.4, language: ron
text: 972x374 px @ 300 dpi, skew +0.00°
wrote tes.txt
Contact: ion.popescuGexample.com
Telefon: 0721 123 456
Bucuresti, 5 noiembrie 2026
Acesta este un registru de test pentru
verificarea ghidului LegatorOCR.
```

The same test page read by the two engines. Both made the same mistake: "@" came out as "(a" with the native engine and as "G" with Tesseract.

## The measured figures (and what they do not say)

8 good pages from a Romanian magazine were measured, against a reference text produced by ABBYY, not corrected by a person. The character error rate, with diacritics (the original measurement, from edition 0.2.0-pre.21; on a new measurement in pre.26 on the same 8 pages, with the same method, the native engine gave 2.76 %; I have not investigated the difference from 2.55 %):

| Engine | Character errors |
|---|---|
| Native | 2.55 % |
| Tesseract (`ron`) | 4.32 % |
| OCRmyPDF (with Tesseract) | 4.62 % |

Read simply: the native engine makes fewer mistakes **on this type of print**. But it was tuned on text from the same sources, so it cannot be said to be better in general. On other documents, especially old or unusual print, it may be weaker than Tesseract. Recognition of whole words is almost equal between the engines (about 92 %).

## Accurate mode and old orthography

- `--accurate` (native engine): each line is read three times (shifted by one pixel up and down) and the probabilities are averaged. It is slower (measured: 18 % to 74 % more time) and helped only on the magazine pages (character error 3.66 % → 3.39 % on 12 pages); on books and brochures it changed nothing measurable. It is off by default.

- The Romanian word list also accepts the orthography of 1953–1993 (`cuvînt`, `romîn`, `sînt`): on five pages from a 1960 textbook, transcribed by hand, correct characters rose from 99.6 % to 99.7 % and words from 96.0 % to 96.3 %. A small difference, on five pages.
- Words are split at wide spaces even when the network did not write a space: the cells of a table no longer come out stuck together (`CantitatePreț`). A table with lines is found as a single table (Excel: `--tables-xlsx`); a table without lines is still found as two tables (left and right), an unsolved problem.

### Tips

- On your own documents, compare the engines on 5–10 pages before the batch: `scripts/compare-ocr-text.py` measures the errors against reference texts that you have.
- Pages under 250 dpi are automatically enlarged before recognition.
- Pages with illustrations or diagrams can produce meaningless text in the area of the image (Tesseract read an illustration as text on one of the measured pages). Check these areas.
- Any OCR can invent or lose words. For deeds, registers or figures, spot-check.

## 8. Outputs and archiving

| What you want | How |
|---|---|
| PDF with searchable text | `--pdf FILE.pdf --ocr` (interface: Export PDF with OCR ticked) |
| PDF for archiving (PDF/A-2b) | `--pdfa`; other parts with `--pdfa-part 1` or `3` |
| Processed images | `-o FOLDER`, with `--format png`, `jpeg` or `tiff`; `--format none` writes no images |
| Plain text of the pages | `--txt FILE.txt` |
| hOCR | the "Also save hOCR" tick box |
| Structure for accessibility | `--tagged ro` (tagged paragraphs). PDF/UA conformance has **not** been validated |
| DjVu copy | `--djvu FILE.djvu` |
| Proof of integrity | `--checksums` and `--archive-package DIR`, then `--verify-sums` |
| Project (settings and corrections) | File → Save project (`.lgr`), or `--save-project`; `--embed-project` hides it in the PDF |

### About PDF/A

The program writes PDF/A-2b with an embedded font and sRGB profile. Conformance was checked on samples, **not** with veraPDF on every output. If an institution requires certified conformance, run your own validator on the final files.

### About ScanTailor projects

`--project FILE.ScanTailor` opens a ScanTailor project: the scan list and the rotation per image are taken over, while the skew, the splitting and the content box are found again. The reader was written from the description of the format and tested on a hand-made project, not on files from every ScanTailor version.

```
$ legatorocr scans/text.png --pdf sp.pdf --save-project pr.lgr --format none --work-area 5,5,80,60 --cut-at 0.5; grep -n
 'work_area\|cut_at' pr.lgr; legatorocr --project pr.lgr --pdf sp2.pdf --format none
project saved: pr.lgr
text_L: 820x374 px @ 300 dpi, skew +0.00°
text_R: 270x354 px @ 300 dpi, skew +0.00°
wrote sp.pdf (2 page(s))
31:work_area = 5,5,80,60
32:cut_at = 0.5
text_L: 820x374 px @ 300 dpi, skew +0.00°
text_R: 270x354 px @ 300 dpi, skew +0.00°
wrote sp2.pdf (2 page(s))
```

--save-project and --project: the project is saved and reopened with the same result (2 pages). Note: in pre.18, --work-area and --cut-at were not kept in the project; fixed in pre.19 (checked: the saved and reopened project gives the same pages).

# 9. Common problems and known limits

| Symptom | What to check |
|---|---|
| The page comes out cut off or with too much white | the content box (section 4); the frame in mm in the "This page" bar |
| A skewed page stays skewed | the maximum skew allowed; set the angle manually |
| The double-page sheet is not cut where it should be | the page splitting mode: "at the middle" or "off"; since pre.18 you can drag a "Cut line" on the original or use `--cut-at 0.45` |
| A text field turns red | the format of the field (section 6): millimetres for zones, fractions 0–1 for guides |
| The OCR text has mistakes | compare the engines (section 7); check the resolution; switch to Tesseract or the other way round |
| The "To check" panel | pages marked as suspect (blank, borderline skew, nothing found, content stuck to the edge); they are not necessarily wrong, but they are worth a look |
| The window does not work on Windows or in RDP | missing OpenGL 2+; try the video card driver |
| The macOS app does not start | it is not signed: right-click → Open or the `xattr` command from section 2 |

## Known limits of edition 0.2.0-pre.32

- Rotation by 90°: on a page with a photograph of ornaments (page 15 of a PDF) a tester on Windows saw the rotation jump from 90° to 180°. I reproduced part of the problem on Linux: in pre.18, with "Detect page lying on its side" ticked (the default), the program wrongly rotated this page by 90°, although the original is upright (the figure in section 4). The threshold in pre.19 was too fragile (I have meanwhile discovered that another page of the same PDF, page 13, was also rotated wrongly). In pre.20 the detector counts only pieces the size of a letter and abstains on pages with pictures or with few words. On 98 book pages (rotated at 0° and 90°) and 5 pages with pictures, the result is identical to pre.18, apart from page 13 (rotated wrongly in pre.18, now upright) and one page with a photo lying on its side, which the detector no longer turns (OSD and manual rotation remain). Checked on the command line, on Linux, on a small corpus. The probable cause of the jump (automatic orientation added on top of the manual rotation) was fixed in pre.18: after a click on rotate the page stays locked, and four clicks give 90°, 180°, 270° and back to 0° (checked in the window, on Linux). What exactly the tester saw on Windows 11 I could not check. If a page with pictures is still rotated wrongly, untick "Detect page lying on its side" or use --no-auto-orient.

- The work area, the movable cut line, the erase and picture zones, the grid of points, Undo and Ctrl+Z, the content box, OCR and PDF export were tried in a real window on Linux, with a simulated mouse (the screenshots in the guide). Resizing a zone from a corner worked in the window. The native file dialogs ("Export PDF…", "Save project", "Open project") worked with the GTK dialog of an xdg portal on Linux; on Windows and macOS I did not try them. If the system has no dialog at all, since pre.21 the window says so instead of doing nothing (you can drag the files into the window or use the command line). Saving a custom profile worked. The effect of the cut line I checked through automated tests, with --cut-at and in the window (a double sheet cut into two unequal pages, 524 and 674 px). For sheets rotated by 90° or 270° (since pre.21) the pages come out one below the other; this is checked only through an automated test, not in the window.

  --save-project: in the published build pre.18, --work-area and --cut-at given on the command line were not kept in the .lgr file (found while checking this guide). Fixed in pre.19. If you use pre.18, give the options again with --project, not only from the project.

- The graphical interface has not been checked by people on Windows, macOS or arm64.
- PDF/UA verification, the CIUS-RO electronic invoice and the time stamp over HTTPS with a real authority have not been validated.
- For a certified PDF/A copy, the validator is yours (for example veraPDF).
- The right to use the program's own OCR model (partly trained on pages with unclarified rights) is an open matter for the author; see `docs/audit/assets-licences.md`.

# 10. How to report a problem

The most useful contribution is a report from a system the author has not tried. It takes five minutes:

1. Download the latest edition for your system.
2. Run `legatorocr --doctor` and copy the whole output.
3. Run a page you can share: `legatorocr PAGE.jpg --ocr --pdf result.pdf -o pages`.
4. Open `result.pdf`: is the page upright and cropped as it should be? Can the text be selected and searched?

Send, as an issue in the `chiuta/LegatorOCR` repository (or by e-mail):

- the system and its version (for example Windows 11 23H2, macOS 15 arm64, Debian 12 arm64);
- the LegatorOCR version;
- the output of the `--doctor` command;
- the command you ran (or the setting in the interface);
- what you expected and what happened (error text or screenshot);
- whether the image can be shared.

Do not send documents with personal data. An "everything worked" report is still useful: it is the only way the author learns that a system works.

Security problems are reported according to `SECURITY.md`, not publicly.